

Benchmarking Explainable AI Methods for ECG Classification Using Perturbation-Based Faithfulness Evaluation

Master Thesis

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Abstract

Electrocardiography (ECG) is one of the primary diagnostic tools for detecting cardiovascular disease, including Myocardial Infarction (MI). Although deep learning models have achieved high performance in automated ECG classification, their limited interpretability remains a major obstacle to clinical adoption. Explainable Artificial Intelligence (XAI) addresses this challenge by providing insight into the features that influence model predictions. However, evaluating whether these explanations faithfully reflect the model's decision-making process remains an open research problem.

This thesis investigates explainable deep learning for ECG analysis using the PTB-XL dataset. Two complementary deep learning architectures, a CNN and a CNN–BiLSTM with Attention, were developed and evaluated for multi-class ECG classification. In a separate binary MI versus Normal sinus rhythm (NORM) classification task, the same architectures were trained to enable a detailed explainability study. Model explanations were generated using Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD).

To quantitatively evaluate explanation quality, a perturbation-based benchmark was proposed in which the ECG regions identified as important by each attribution method are progressively perturbed while monitoring the resulting degradation in prediction performance. The evaluation considers both zero-out and interpolation perturbation strategies at segment- and timestep-level granularities and is complemented by statistical analyses based on perturbation Area Under the Curve (AUC), bootstrap confidence intervals, Wilcoxon signed-rank tests, and Bland–Altman agreement analysis.

The experimental results show that attribution-guided perturbation consistently produces greater performance degradation than random perturbation, indicating that the attribution explanations identify signal regions that are important for the model's predictions. Among the evaluated attribution methods, Deep Taylor Decomposition generally exhibited the highest attribution faithfulness under the evaluated perturbation benchmark, while the CNN–BiLSTM with Attention model shows higher attribution faithfulness than the CNN. Overall, the proposed evaluation framework provides a systematic and statistically grounded approach for assessing explanation quality in deep learning models for ECG analysis.

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1

Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, accounting for an estimated 17.9 million lives each year, according to the World Health Organization. Among the most common and life-threatening of these conditions is Myocardial Infarction (MI), commonly known as a heart attack, which occurs when blood flow to a part of the heart muscle is suddenly blocked. Therefore, early and accurate detection of MI is critical, as timely clinical intervention can significantly reduce mortality and long-term cardiac damage [1].

The 12-lead Electrocardiogram (ECG) is the primary non-invasive tool used in clinical practice for the diagnosis of cardiovascular diseases. It records the electrical activity of the heart from twelve different perspectives simultaneously, capturing characteristic patterns associated with various pathologies. Despite its widespread use, accurate interpretation of ECG signals requires significant clinical expertise, and inter-observer variability remains a known limitation in practice. This motivates the development of automated, data-driven approaches capable of supporting clinicians in ECG-based diagnosis.

Also In recent years, deep learning models have achieved remarkable performance in automated ECG classification tasks, often matching or exceeding the accuracy of trained cardiologists on large benchmark datasets. However, the black-box nature of these models poses a fundamental challenge to clinical adoption. In safety-critical medical settings, it is not sufficient for a model to produce a correct prediction, and clinicians must be able to understand why the model reached a particular decision. This need has led to the field of Explainable Artificial Intelligence (XAI), which develops methods that make the reasoning of deep learning models transparent and interpretable [2].

In this thesis, we investigate explainable artificial intelligence for deep learning-based ECG classification using the PTB-XL dataset. We first develop and evaluate two complementary deep learning architectures for multi-class ECG classification. Building on these classification models, we subsequently focus on binary Myocardial Infarction (MI) versus Normal sinus rhythm (NORM) classification to perform a detailed explanation analysis. The evaluated architectures include a deep Convolutional Neural Network (CNN) and a hybrid CNN–BiLSTM with Attention model. Although the CNN is designed to learn hierarchical local morphological features from ECG waveforms, the CNN–BiLSTM with Attention model also captures long-range temporal dependencies through recurrent

processing and attention, enabling a comparison between two complementary approaches to ECG classification.

To explain model predictions, we applied three attribution methods: Integrated Gradients (IG), SmoothGrad-IG, a smoothing extension of Integrated Gradients, and Deep Taylor Decomposition (DTD). Their faithfulness is evaluated using a perturbation-based benchmark in which the most relevant ECG regions identified by each method are progressively perturbed and the resulting degradation in prediction performance is measured. Two perturbation strategies, zero-out and interpolation toward a synthetic healthy ECG baseline, are investigated at both segment- and timestep-level granularities, providing a comprehensive perturbation-based evaluation of attribution quality.

To complement the perturbation-based evaluation, we performed a Bland–Altman agreement analysis to assess the consistency of perturbation AUC measurements across different attribution methods, model architectures, and perturbation strategies. This analysis provides a sample-level assessment of agreement and complements the population-level evaluation obtained from the perturbation benchmark.

Contributions

The main contributions of this thesis are as follows.

- We develop and evaluate deep learning models for both multi-class ECG classification and binary Myocardial Infarction (MI) detection on the PTB-XL dataset, establishing the foundation for the subsequent explainability analysis.
- We implement and compare three attribution methods, Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD), to explain model predictions on 12-lead ECG signals.
- We develop a perturbation-based evaluation framework to comprehensively assess the faithfulness of attribution methods across different model architectures, perturbation strategies, and perturbation granularities.
- We perform comprehensive statistical analyses, including Bland–Altman agreement analysis, to compare attribution methods, perturbation strategies, and model architectures.

Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 presents the clinical and technical background on electrocardiography and Explainable Artificial Intelligence.

Chapter 3 describes the proposed methodology. Chapter 4 presents the experimental results, followed by a discussion in Chapter 5. Finally, Chapter 6 concludes the thesis and outlines directions for future work.

2

Background

2.1 Cardiovascular Disease

Cardiovascular diseases (CVDs) comprise a broad group of disorders affecting the heart and vascular system, including coronary artery disease, heart failure, cerebrovascular disease, peripheral arterial disease, and cardiomyopathies. Despite advances in prevention, diagnosis, and treatment, CVDs remain a leading cause of mortality worldwide and place a substantial burden on healthcare systems. Findings from the Global Burden of Disease (GBD) Study indicate that both the number of CVD-related deaths and the overall prevalence of cardiovascular disease have increased over recent decades. This trend is largely associated with population growth, population ageing, and the continued presence of modifiable cardiovascular risk factors. CVDs therefore remain a major global public health challenge, underscoring the need for improved approaches to early diagnosis and clinical management [3]

Among cardiovascular diseases, ischaemic heart disease is the leading cause of cardiovascular mortality and disability worldwide. Its most acute manifestation is myocardial infarction (MI), which occurs when coronary blood flow is abruptly reduced or interrupted, most commonly following rupture or erosion of an atherosclerotic plaque with subsequent thrombus formation. The resulting oxygen deprivation causes myocardial injury that may rapidly progress to irreversible tissue necrosis if coronary perfusion is not promptly restored. Because clinical outcomes depend on the time to reperfusion, rapid and accurate diagnosis is essential to reduce myocardial damage, preserve cardiac function, and

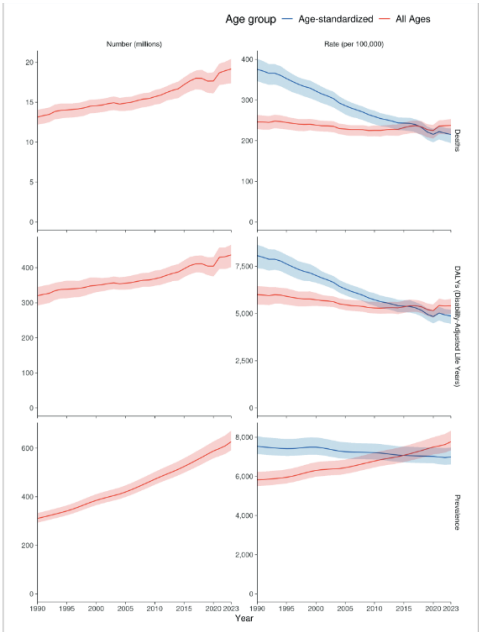


Figure 1: Global CVD deaths, DALYs, and prevalence (1990–2023): rising absolute counts but declining age-standardised rates.[3].

improve patient survival. Current clinical guidelines therefore emphasize immediate evaluation and early therapeutic intervention for patients with suspected acute myocardial infarction [4].

The 12-lead electrocardiogram (ECG) is the primary diagnostic tool for patients with suspected myocardial infarction because it provides a rapid and non-invasive assessment of the heart's electrical activity. Characteristic ECG changes reflect the electrical effects of myocardial ischaemia and help detect and localize myocardial injury at an early stage. Because of its wide availability, low cost, and established clinical value, the ECG remains the first-line investigation for patients with suspected acute coronary syndromes [5].

2.2 Electrocardiogram (ECG)

The electrocardiogram (ECG) is a non-invasive technique that records the electrical activity of the heart using electrodes placed on the body surface.[6] Each heartbeat is initiated by an electrical impulse originating from the sinoatrial (SA) node, which propagates through the atria, passes the atrioventricular (AV) node, and is then conducted through the His–Purkinje system to activate the ventricular myocardium. This coordinated sequence of electrical conduction generates electrical potentials that can be detected at the body surface and recorded as an ECG signal [7].

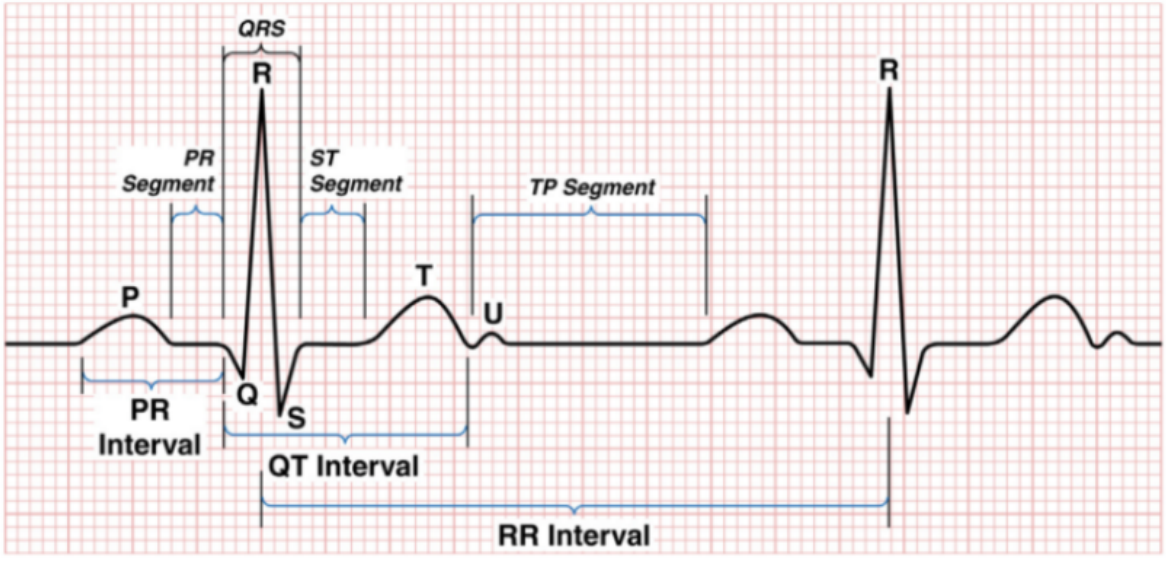


Figure 2: Main ECG components, showing standard waveforms, intervals, and segments used for clinical interpretation.[8]

Each cardiac cycle is represented by a sequence of characteristic waveforms and intervals, as illustrated in Figure 2. The main components are the P wave, the QRS complex, the ST segment, and the T wave, which correspond to atrial depolarisation, ventricular

depolarisation, early ventricular repolarisation, and ventricular repolarisation, respectively. Alterations in the morphology or timing of these components may indicate abnormalities in cardiac conduction or myocardial injury [9].

Of these components, the QRS complex, ST segment, and T wave are particularly important for identifying myocardial ischaemia and infarction, as these conditions produce characteristic abnormalities in the ECG waveform. These changes are central to the electrocardiographic diagnosis of myocardial infarction and are discussed in the following section [10].

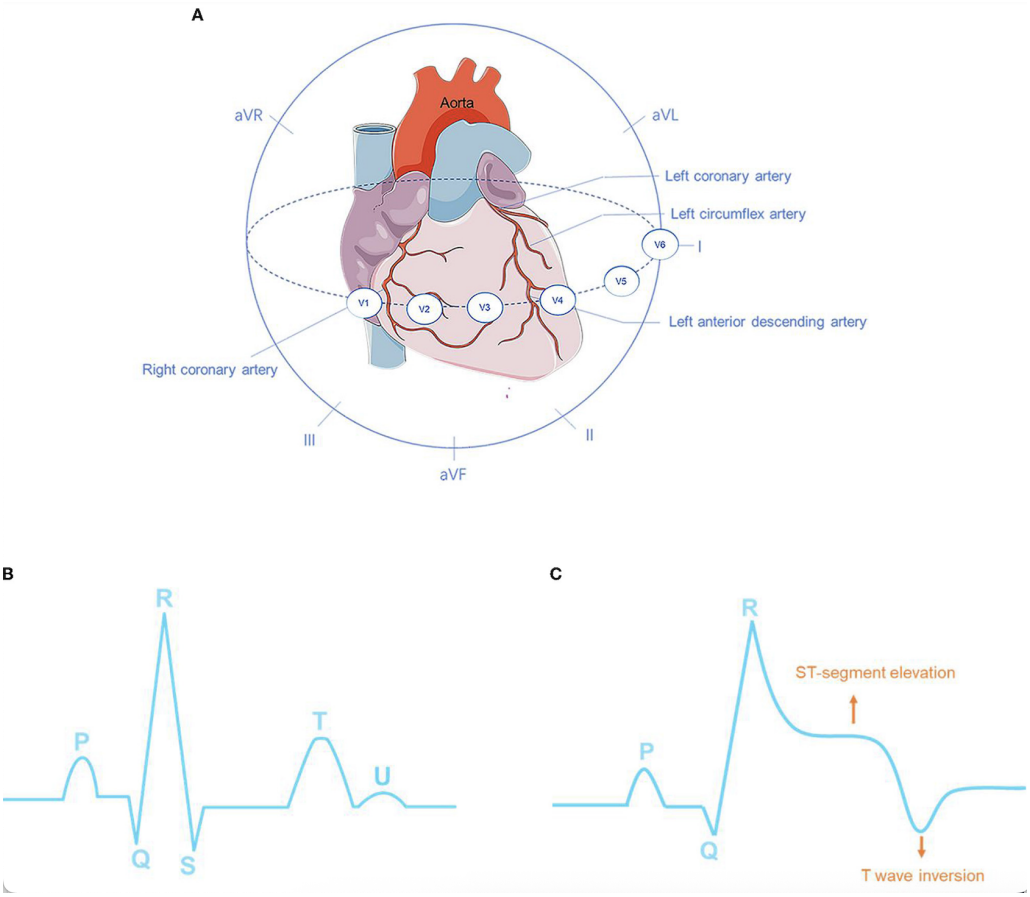


Figure 3: (A) Standard 12-lead ECG electrode placement and its anatomical relationship to the coronary arteries. (B) Normal ECG waveform showing the P wave, QRS complex, and T wave. (C) Representative ECG changes associated with acute myocardial infarction, including ST-segment elevation and T-wave inversion. Source: [11].

The standard clinical ECG consists of twelve leads derived from ten electrodes placed on the limbs and chest. Each lead records cardiac electrical activity from a different spatial perspective, allowing clinicians to assess both the temporal and spatial characteristics of cardiac depolarisation and repolarisation. This multi-lead configuration provides complementary information that improves the detection and localisation of cardiac abnormalities compared with single-lead recordings [12].

The twelve leads are grouped into three categories: the three bipolar limb leads (I, II, and

III), the three augmented limb leads (aVR, aVL, and aVF), and the six precordial leads (V1–V6)[10]. Together, they provide views of the heart in both the frontal and horizontal planes. As shown in Figure 3A, each lead observes the heart from a different anatomical perspective, allowing abnormalities to be localised to specific myocardial regions supplied by different coronary arteries. This spatial information is essential for identifying the location of myocardial infarction [10].

Myocardial infarction (MI) alters the normal electrical activity of the heart because myocardial ischaemia and necrosis disrupt impulse propagation through the affected tissue. As a result, characteristic ECG changes appear, making the ECG an essential tool for the early diagnosis of acute myocardial infarction.

Figure 3C illustrates representative ECG changes associated with acute myocardial infarction, alongside a normal ECG waveform shown in Figure 3B. Whereas a normal cardiac cycle consists of distinct P waves, QRS complexes, and T waves, acute myocardial infarction is commonly associated with ST-segment elevation or depression, T-wave inversion, and, in later stages, the development of pathological Q waves. The type and distribution of these abnormalities depend on the affected coronary artery and the stage of infarction.

These characteristic ECG abnormalities form the basis of both clinical interpretation and automated ECG analysis. Their reliable detection is fundamental for computer-aided diagnosis and has motivated the development of machine learning and deep learning methods for automated ECG interpretation [11].

2.3 Deep Learning

Artificial intelligence (AI) has become an important tool in medical data analysis by enabling the automated detection of complex patterns in large datasets. Machine learning (ML), a subset of AI, learns predictive patterns directly from data, whereas deep learning (DL) uses multi-layer neural networks to automatically learn hierarchical feature representations. Although deep learning has gained widespread attention in recent years, its underlying concepts date back several decades [13]. Multilayer non-linear networks were studied during the 1960s and 1970s, and backpropagation-based learning methods also originated during this period . Unlike traditional machine learning approaches, which often rely on handcrafted features, deep learning models can learn informative representations directly from raw ECG signals, making them particularly well suited to automated ECG interpretation.[14]

An artificial neural network consists of interconnected layers of computational units, known as neurons. The input layer receives the input data, such as an ECG waveform, an image, or a feature vector, and passes the information through one or more hidden layers before

reaching the output layer, which generates the final prediction. Each neuron computes a weighted sum of its inputs, applies a non-linear activation function, and passes the resulting output to the next layer. The activation function enables the network to model non-linear relationships; without it, multiple linear transformations would be equivalent to a single linear transformation regardless of the network depth.

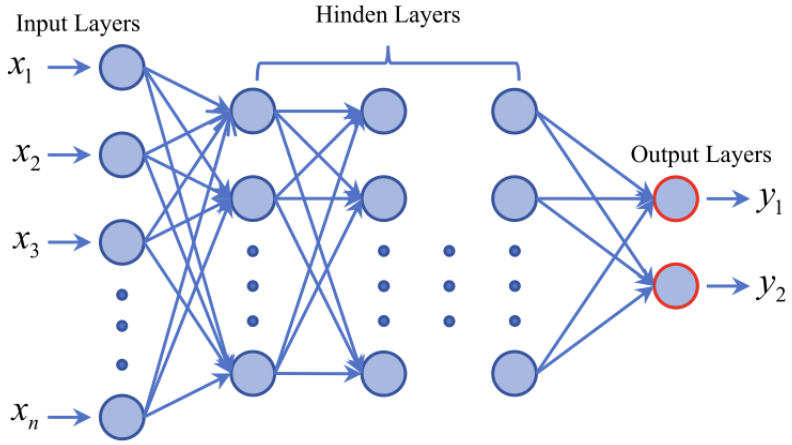


Figure 4: Architecture of a feedforward neural network consisting of an input layer, multiple hidden layers, and an output layer. Source:[15]

Training begins with forward propagation, during which an input passes through the network to produce a prediction. A loss function then measures the discrepancy between the predicted output and the ground-truth label. The gradient of the loss with respect to each model parameter is subsequently computed using backpropagation, an efficient application of the chain rule through the computational graph . These gradients are used by a gradient-based optimisation algorithm, typically a variant of stochastic gradient descent, to iteratively update the network parameters until convergence.[16]

Among deep learning architectures, Convolutional Neural Networks (CNNs) are particularly well suited to ECG analysis because they automatically learn local temporal features directly from raw signals. The following section therefore introduces the fundamental principles of CNNs. [14]

2.4 Convolutional Neural Networks

Although originally developed for image recognition, Convolutional Neural Networks (CNNs) have also proven highly effective for analysing one-dimensional biomedical signals such as ECG recordings [17]. Their ability to automatically learn hierarchical feature representations directly from raw input eliminates the need for handcrafted feature extraction and enables the identification of complex waveform patterns associated with cardiovascular diseases.

Unlike fully connected neural networks, CNNs exploit local connectivity and parameter sharing, allowing the same learnable filter to be applied across different regions of the input while substantially reducing the number of trainable parameters. The core operation of a CNN is convolution, in which a learnable filter, or kernel, slides across the input signal and computes a dot product at each position. This process produces a feature map that indicates where the learned pattern is present and how strongly it is activated [18].

A typical CNN consists of a sequence of convolutional layers (Figure 5), followed by non-linear activation functions and pooling layers. The convolutional layers extract local features from the input signal, while deeper layers progressively combine these low-level features into increasingly abstract representations.

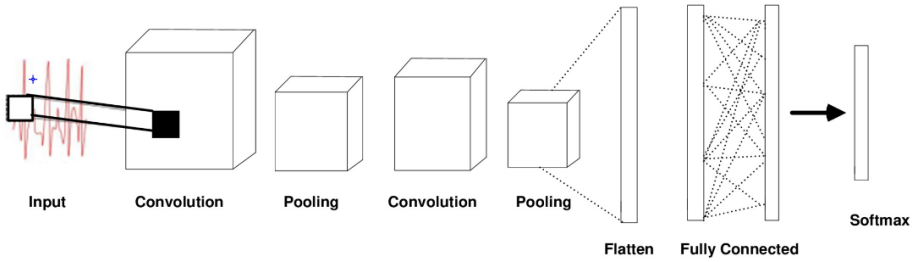


Figure 5: Typical 1D CNN architecture for time-series classification, consisting of convolutional and pooling layers for feature extraction, followed by fully connected layers and a Softmax output layer. Source: [19].

The one-dimensional convolution operation is defined as

$$y_i = \sum_{k=0}^{K-1} x_{i+k} w_k + b, \tag{1}$$

where x denotes the input ECG signal, w represents the learnable convolution kernel of length K, b is the bias term, and y_i is the output feature at position i. During training, the kernel parameters are updated through backpropagation, allowing the network to learn filters that are useful for the classification task.

The kernel is applied repeatedly across the input signal using a predefined stride, which determines how many samples the kernel moves after each convolution. The output length is given by

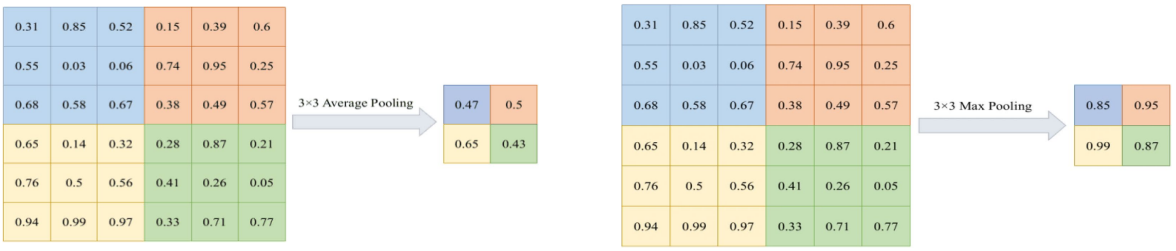
$$L_{\text{out}} = \left\lfloor \frac{L_{\text{in}} - K + 2P}{S} \right\rfloor + 1, \tag{2}$$

where L_{in} is the input length, K is the kernel size, P denotes the padding, and S is the stride. A larger stride reduces the spatial resolution of the feature maps, whereas padding preserves information near the signal boundaries and helps control the output dimensions.

Following each convolution, a non-linear activation function is applied. In this work, the Rectified Linear Unit (ReLU) is used and is defined as

$$f(x) = \max(0, x). \tag{3}$$

ReLU introduces non-linearity into the network while remaining computationally efficient. It also helps mitigate the vanishing gradient problem during training.



(a) Average Pooling (b) Max Pooling

Figure 6: Comparison of 3×3 average pooling (left) and max pooling (right) [20]

Pooling layers are commonly used after convolutional layers to reduce the dimensionality of feature maps and improve computational efficiency. Max pooling (see Figure 6B) retains the maximum activation within each pooling region and is expressed as

$$y_i = \max_{j \in \Omega_i} x_j, \tag{4}$$

where Ω_i denotes the pooling window associated with output position i. By retaining only the strongest activation within each pooling region, max pooling reduces memory requirements, improves robustness to small shifts in the input signal, and helps reduce the risk of overfitting.

Average pooling (see Figure 6A), in contrast, computes the mean of all activations within the pooling window and is defined as

$$y_i = \frac{1}{|\Omega_i|} \sum_{j \in \Omega_i} x_j, \tag{5}$$

where $|\Omega_i|$ denotes the number of elements in the pooling window Ω_i. In this work, a global average pooling layer is applied after the final convolutional block to reduce each feature map to a single value before it is passed to the fully connected layer.

To improve training stability, Batch Normalization is commonly applied between convolution and activation layers. It normalizes intermediate activations according to

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \varepsilon}}, \tag{6}$$

followed by a learnable scaling and shifting operation,

$$y = \gamma \hat{x} + \beta, \tag{7}$$

where μ and σ^2 denote the batch mean and variance, respectively, ε is a small constant introduced for numerical stability, and γ and β are learnable parameters.

After several convolutional blocks, the extracted features are flattened and passed to one or more fully connected layers, where the learned representations are integrated for classification. The fully connected operation is given by

$$z = Wx + b, \tag{8}$$

where W is the weight matrix, x is the input feature vector, and b is the bias term.

Finally, the network converts the output logits into class probabilities using the Softmax function,

$$P(y = i) = \frac{\exp(z_i)}{\sum_{j=1}^{C} \exp(z_j)}, \tag{9}$$

where C denotes the number of output classes.

The network parameters are optimized by minimizing the categorical cross-entropy loss,

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \tag{10}$$

where y_i is the ground-truth label and $\hat{y}_i$ is the predicted probability for class i. During training, the gradients of this loss are propagated backward through the network using the backpropagation algorithm, allowing the network parameters to be updated iteratively until the model converges.

In the context of ECG analysis, a 12-lead recording can be represented as a matrix $X \in \mathbb{R}^{12 \times T}$, where each row corresponds to one lead and each column corresponds to a time step. For the PTB-XL dataset used in this thesis, $T = 1000$ samples at a sampling frequency of 100 Hz. A one-dimensional convolutional kernel slides along the temporal axis

while processing all leads simultaneously, enabling the network to learn local waveform patterns, such as the QRS complex, ST segment, and T wave, directly from the raw ECG signal [21].

2.5 Convolutional Neural Network–Bidirectional Long Short-Term Memory Networks

While CNNs are highly effective at extracting local morphological features, their receptive field is inherently limited [22]. ECG recordings also contain long-range temporal dependencies, including relationships between consecutive heartbeats and rhythm variations over time. Capturing these sequential patterns often requires a model that can retain information across longer time intervals. For this reason, CNNs are frequently combined with Long Short-Term Memory (LSTM) networks. [22]

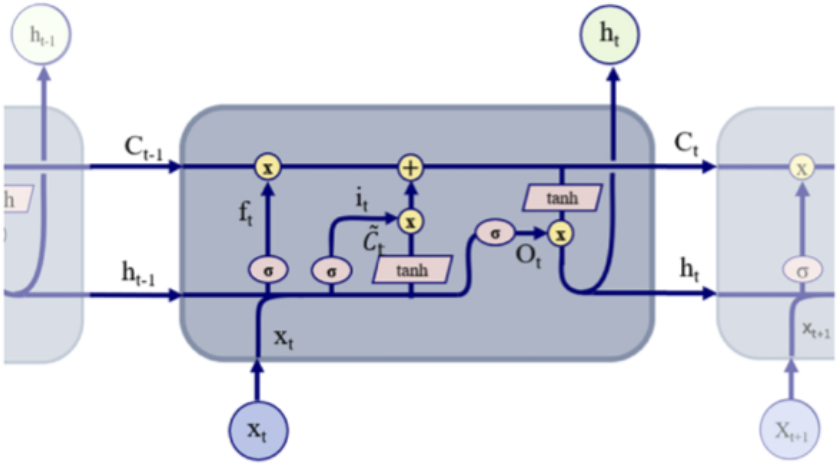


Figure 7: Internal structure of an LSTM cell, showing the forget gate (f_t), input gate (i_t), candidate cell state ($\tilde{C}_t$), cell update (C_t), and output gate (o_t), which together regulate the flow of information across time steps. Source: [23].

LSTM is a specialised type of recurrent neural network (RNN) designed to address the vanishing gradient problem [24] commonly encountered in conventional RNNs. As shown in Figure 7, each LSTM cell maintains an internal memory state, C_t, together with three gating mechanisms that regulate how information is retained, updated, and propagated through the sequence. The gating equations of an LSTM cell are given by [25]

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)\,, \tag{11a}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)\,, \tag{11b}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)\,, \tag{11c}$$

where f_t, i_t, and o_t denote the forget, input, and output gates, respectively, σ is the sigmoid activation function, h_{t-1} is the previous hidden state, and x_t is the current input. The candidate memory state and the updated cell state are computed as

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)\,, \tag{12a}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \tag{12b}$$

where $\odot$ denotes element-wise multiplication. The hidden state passed to the next time step is then computed as

$$h_t = o_t \odot \tanh(C_t). \tag{13}$$

In this thesis, a Bidirectional LSTM (BiLSTM) is used to capture contextual information from both past and future observations. Instead of processing the sequence in only one direction, a BiLSTM analyses it in both forward and backward directions [26]. The hidden representations from the two passes are then concatenated to form the final representation,

$$h_t = \left[\, \overrightarrow{h_t}\,;\, \overleftarrow{h_t} \,\right]. \tag{14}$$

An attention mechanism is subsequently applied to assign different importance weights [27] to the hidden states generated by the BiLSTM. The attention score, attention weight, and context vector are computed as

$$e_t = v^\top \tanh(W h_t + b)\,, \tag{15a}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)}, \tag{15b}$$

$$c = \sum_t \alpha_t\, h_t, \tag{15c}$$

where α_t represents the attention weight assigned to the hidden state at time step t, and c is the resulting context vector used for the final classification.

Combining CNNs, BiLSTMs, and attention mechanisms (Figure 8) enables the model to capture local waveform morphology, long-range temporal dependencies, and the most diagnostically relevant regions of the ECG signal within a unified architecture. The CNN first processes the raw ECG signal and extracts a sequence of local feature representations, transforming the high-dimensional waveform into a more compact sequence of learned features. These representations are then passed to the BiLSTM, which models temporal relationships between successive feature vectors while preserving contextual information over time. Finally, the attention mechanism assigns higher weights to the most informative temporal features [27] before the final classification stage. By combining these three components, the architecture learns complementary spatial and temporal representations that are well suited for ECG classification [28].

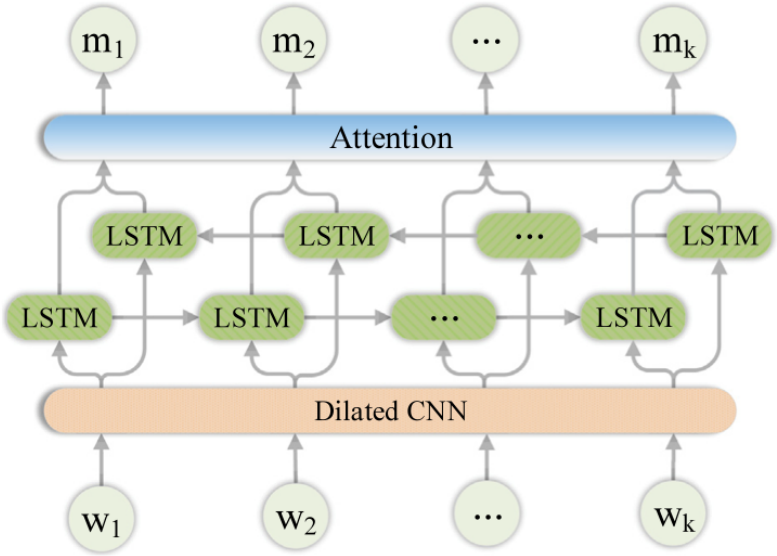


Figure 8: Architecture of a CNN–BiLSTM model with an attention mechanism. Source: [29].

For ECG classification, these components serve complementary roles: the CNN extracts local waveform features, the BiLSTM captures longer-range temporal dependencies, and the attention mechanism emphasises the most informative time steps before classification. [18].

2.6 Explainable Artificial Intelligence (XAI)

Deep learning models have achieved remarkable performance across a wide range of applications, but their increasing complexity has made them more difficult to interpret. Since the relationship between the input and output cannot be inferred directly from the network architecture or its learned parameters, understanding individual predictions

is often challenging. This lack of transparency is particularly important in high-stakes domains such as healthcare [30]. Explainable Artificial Intelligence (XAI) aims to improve the transparency and interpretability of deep learning models by providing methods that support model verification, debugging, and more trustworthy deployment [31].

Although highly accurate models are often considered less interpretable, this trade-off is not universal [30].

Compared with simpler machine learning models, such as decision trees or linear classifiers, the decision-making process of deep neural networks is considerably more difficult to inspect directly. As a result, numerous post-hoc explanation methods have been developed to interpret trained models without modifying their architecture or predictive performance.

In healthcare, limited model interpretability can reduce trust in automated clinical decision-support systems. A model may achieve high performance on benchmark datasets while relying on spurious correlations that do not generalise to real-world clinical settings. Without understanding the basis of a prediction, clinicians cannot easily assess its reliability, regulators face greater challenges when auditing the model, and developers may find it difficult to identify potential failures before clinical deployment [32]. Consequently, explainability has become an important consideration for the safe and responsible use of AI in healthcare, although the type and level of explanation required depend on the clinical application and the needs of different users [33, 32].

As illustrated in Figure 9, XAI methods are commonly categorised according to several dimensions. One distinction is between *intrinsic* and *post-hoc* interpretability. Intrinsic models are inherently transparent and explainable by design, whereas post-hoc methods provide explanations for the predictions of an already trained black-box model [34, 33]. Another distinction concerns the scope of the explanation: *global* methods describe the overall behaviour of a model across the input space, whereas *local* methods explain individual predictions by identifying the features that contributed most to a specific decision [34, 35].

XAI methods can also be categorised as *model-agnostic* or *model-specific*. Model-agnostic approaches can be applied to a wide range of model architectures, whereas model-specific methods exploit the internal structure of a particular model to generate more targeted explanations [30].For deep learning models applied to ECG analysis, post-hoc local explanation methods are particularly relevant: they clarify individual predictions from otherwise difficult-to-interpret models, which aligns with clinicians' need to understand the rationale behind each prediction [33].

Within post-hoc XAI, explanation methods can be broadly divided into several families. Gradient-based methods estimate feature importance from the model gradients, perturbation-based methods evaluate how changes to the input affect the model output, and propagation-based methods redistribute the model output back to the input using

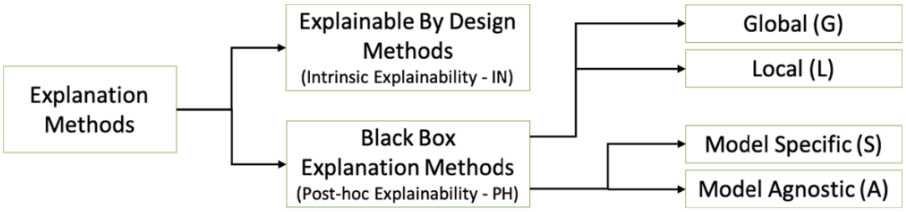


Figure 9: Taxonomy of explanation methods, distinguishing between intrinsic (Explainable by Design) and post-hoc (Black Box) approaches, further categorised by scope (global vs. local) and model dependency (model-specific vs. model-agnostic). Source: [30]

layer-wise relevance propagation rules [36]. In this thesis, the perturbation principle is adopted to evaluate attribution methods rather than to generate explanations. Since these approaches rely on different underlying principles, they may produce different explanations for the same prediction, making the evaluation of explanation quality an ongoing research challenge [37, 32]. Integrated Gradients and SmoothGrad-IG are employed as gradient-based attribution methods, whereas Deep Taylor Decomposition is used as a propagation-based attribution method to explain ECG classification decisions.

2.7 Integrated Gradients (IG)

Integrated Gradients (IG) was introduced by Sundararajan et al. as an axiomatic attribution method for deep neural networks . It was proposed to address a key limitation of standard gradient-based saliency methods, whose gradients reflect only the local sensitivity of the model and may be close to zero even when a feature has made an important contribution to the prediction [38]. Instead of evaluating the gradient only at the input, IG accumulates gradients along a straight-line path from a reference input, referred to as the *baseline* $\mathbf{x}'$, to the actual input $\mathbf{x}$ [38]. As illustrated in Figure 10, IG follows the straight-line path P_2 between the baseline and the input, rather than alternative non-linear paths such as P_1 and P_3. The attribution assigned to the i-th input feature is

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - x_i') \times \int_0^1 \frac{\partial F\Big(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')\Big)}{\partial x_i}\, d\alpha, \tag{16}$$

where $\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')$ represents the linear interpolation between the baseline and the input for $\alpha \in [0, 1]$, and $(x_i - x_i')$ scales the accumulated gradient by the total displacement of the feature. The baseline serves as a reference input, typically a zero-valued signal, and all attributions are interpreted relative to this choice [38, 39]. Integrated Gradients satisfies three desirable axioms: sensitivity, which assigns non-zero attribution to features that influence the prediction; implementation invariance, which ensures that functionally equivalent networks produce identical attributions; and completeness, which states that the sum of all feature attributions equals the difference between the model output for the

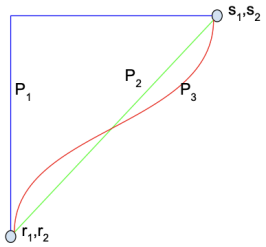


Figure 10: Three interpolation paths between a baseline (r_1, r_2) and an input (s_1, s_2). The straight-line path P_2 corresponds to the interpolation path used by Integrated Gradients. Source: [38].

input and the baseline:

$$\sum_i \mathrm{IG}_i(\mathbf{x}) = F(\mathbf{x}) - F(\mathbf{x}'). \tag{17}$$

In practice, the integral is approximated numerically using a finite number of interpolation points, typically around 20 to 300 steps [39, 40]. A limitation of IG is that the resulting attribution depends on the choice of baseline, and inappropriate baseline selection can weaken interpretability . Because attribution maps can also be visually noisy, smoothing approaches such as SmoothGrad are often used to improve visualization quality [40].

2.8 SmoothGrad

Gradient-based attribution methods often produce visually noisy attribution maps because they reflect only the local sensitivity of the model at a single point in the input space. As a result, the resulting explanations may appear fragmented and difficult to interpret. SmoothGrad, introduced by Smilkov , addresses this limitation by applying Monte Carlo averaging. Instead of generating a single attribution map for the original input, it creates N noisy copies by adding independently sampled Gaussian noise and averages the resulting attribution maps to obtain the final explanation. Formally,

$$\hat{A}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} A\Big(\mathbf{x} + \varepsilon_i\Big), \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \tag{18}$$

where $A(\cdot)$ denotes any gradient-based attribution function, ε_i is a Gaussian noise vector, and N is the number of noisy samples. Features that consistently influence the prediction tend to retain high attribution values across different noisy inputs, whereas local gradient fluctuations are reduced through averaging. Since $A(\cdot)$ can represent any gradient-based attribution method, SmoothGrad acts as a general enhancement technique. In this thesis, it is applied to Integrated Gradients, resulting in the SmoothGrad-IG method [41].

Figure 11 illustrates the influence of the noise level σ on attribution quality. At low noise

levels the attribution maps remain fragmented, become smoother and more coherent at moderate noise levels, and gradually deteriorate when the noise becomes excessively large.

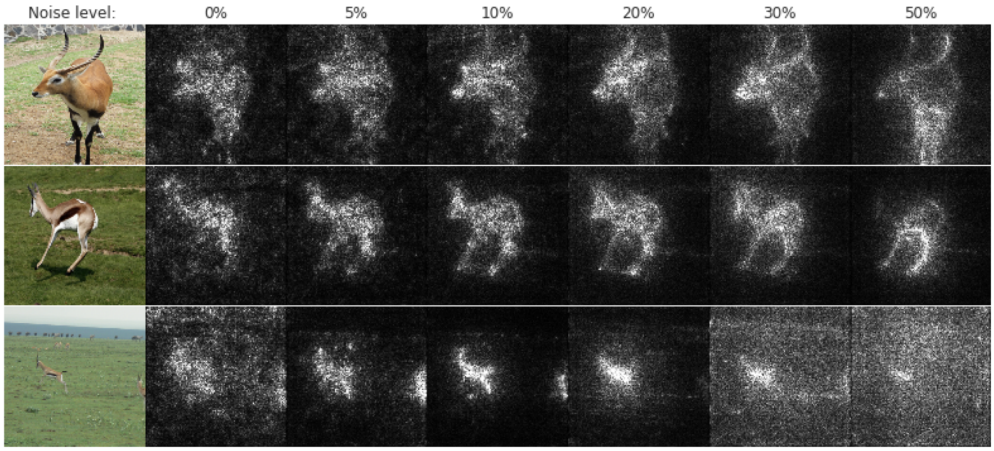


Figure 11: Attribution maps produced by SmoothGrad at increasing noise levels for three ImageNet images. At low values of σ, the maps remain fragmented, become more coherent at moderate noise levels (10–20%), and deteriorate again when the noise is too large. Source: [41].

The two main hyperparameters of SmoothGrad are σ and N. The noise level σ is typically set to 10–20% of the input value range, while N usually ranges from several tens to several hundred noisy samples [42, 41]. Because a separate forward and backward pass is required for each noisy sample, the computational cost increases approximately linearly with N. Although SmoothGrad generally produces visually smoother attribution maps, improvements in robustness and attribution quality depend on the underlying model and are not consistently observed across different architectures [42, 43].

Like Integrated Gradients, SmoothGrad-IG is a gradient-based attribution method. In contrast, propagation-based approaches explain predictions by redistributing the model output backwards through the network according to layer-wise conservation rules. The following subsection introduces Deep Taylor Decomposition, one of the most widely used methods in this family.

2.9 Deep Taylor Decomposition

While gradient-based methods analyse a model by measuring input sensitivity, Deep Taylor Decomposition (DTD), follows a fundamentally different approach. Instead of relying on gradients, DTD explains a prediction by propagating the model output backwards through the network, assigning each input feature a *relevance score* that reflects its contribution to the final decision [44].

The central principle of DTD is relevance conservation, which requires the sum of all input

relevance scores to equal the model output $f(\mathbf{x})$:

$$f_c(\mathbf{x}) = \sum_i R_i \tag{19}$$

This conservation property holds at every layer of the network, not only at the input. If $R_i^{(l)}$ and $R_j^{(l+1)}$ denote the relevance scores at adjacent layers l and $l+1$, then

$$\sum_i R_i^{(l)} = \sum_j R_j^{(l+1)} \tag{20}$$

DTD is closely related to Layer-wise Relevance Propagation (LRP). Both methods explain predictions by propagating relevance scores backwards through the network while preserving the total amount of relevance across layers. In particular, DTD provides the theoretical foundation for several widely used LRP propagation rules, including the z-rule, ϵ-rule, and $\alpha\beta$-rule, by deriving them from first-order Taylor expansions around suitable root points [44, 45].

The mathematical foundation of DTD is a first-order Taylor expansion around a *root point* $\tilde{\mathbf{x}}$, defined as a point in the input space where the neuron's activation becomes zero,

$$h(\tilde{\mathbf{x}}) = 0. \tag{21}$$

The root point serves as a reference around which the neuron activation is locally approximated using a first-order Taylor expansion. This approximation provides the basis for redistributing relevance to the input features. Expanding the neuron activation around $\tilde{\mathbf{x}}$ eliminates the constant term of the Taylor series, allowing the prediction to be expressed as a sum of feature-wise contributions. The resulting first-order Taylor approximation is

$$h(\mathbf{x}) \approx h(\tilde{\mathbf{x}}) + \sum_i \left.\frac{\partial h}{\partial x_i}\right|_{\tilde{\mathbf{x}}} (x_i - \tilde{x}_i), \tag{22}$$

and since $h(\tilde{\mathbf{x}}) = 0$, it simplifies to

$$h(\mathbf{x}) \approx \sum_i \frac{\partial h(\tilde{\mathbf{x}})}{\partial x_i}(x_i - \tilde{x}_i). \tag{23}$$

Each term represents the contribution of an individual input feature to the neuron's activation. The contribution is determined by the gradient evaluated at the root point and scaled by the displacement from the root point. Figure 12 illustrates this process, where

the gradient and the displacement are computed separately and combined element-wise to produce the final relevance map $R(\mathbf{x})$.

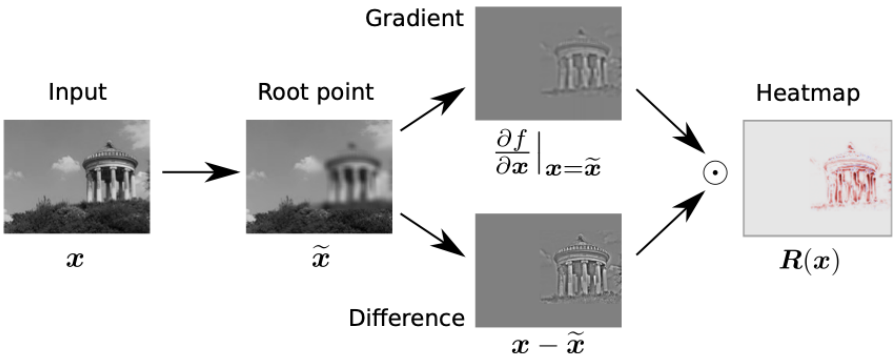


Figure 12: Construction of a DTD relevance map from the input $\mathbf{x}$ and the root point $\tilde{\mathbf{x}}$. The gradient evaluated at the root point and the displacement $(\mathbf{x} - \tilde{\mathbf{x}})$ are combined element-wise ($\odot$) to produce the final relevance map $R(\mathbf{x})$. Positive relevance is shown in red. Source [46]

In practice, explicitly computing a suitable root point for every neuron is computationally intractable. Instead, DTD derives closed-form propagation rules that implicitly correspond to different choices of the root point while preserving relevance conservation.

In the following, let $x_i^{(l)}$ denote the activation of neuron i in layer l, and let w_{ij} denote the weight connecting neuron i in layer l to neuron j in layer $l+1$. The pre-activation of neuron j is defined as

$$z_j = \sum_i x_i^{(l)} w_{ij} + b_j, \tag{24}$$

where b_j is the bias term. The most commonly used propagation rules are described below.

z-rule The z-rule redistributes relevance in proportion to each neuron's contribution:

$$R_i^{(l)} = \sum_j \frac{x_i^{(l)} w_{ij}}{z_j} R_j^{(l+1)} \tag{25}$$

ε-rule The ε-rule introduces a stabilising term to avoid numerical instability when the denominator is close to zero:

$$R_i^{(l)} = \sum_j \frac{x_i^{(l)} w_{ij}}{z_j + \varepsilon \cdot \text{sign}(z_j)} R_j^{(l+1)} \tag{26}$$

$\alpha\beta$-rule The $\alpha\beta$-rule separates positive and negative weight contributions, where the constraint $\alpha - \beta = 1$ preserves relevance conservation:

$$R_i^{(l)} = \sum_j \left[\alpha \cdot \frac{x_i^{(l)} w_{ij}^+}{z_j^+} - \beta \cdot \frac{x_i^{(l)} w_{ij}^-}{z_j^-} \right] R_j^{(l+1)} \tag{27}$$

where

$$w_{ij}^+ = \max(0, w_{ij}), \qquad w_{ij}^- = \min(0, w_{ij}),$$

and

$$z_j^+ = \sum_i x_i^{(l)} w_{ij}^+, \qquad z_j^- = \sum_i x_i^{(l)} w_{ij}^-.$$

The condition $\alpha - \beta = 1$ ensures that the total relevance is conserved across layers [44, 47].

DTD produces class-specific attribution maps because relevance propagation always starts from the score of a particular output class. Compared with raw gradient-based methods, DTD often generates attribution maps that are more stable and easier to interpret, although their characteristics depend on both the network architecture and the propagation rule employed. One limitation of DTD is its sensitivity to the choice of propagation rule, as different rules can produce different relevance distributions. In addition, extending DTD to recurrent architectures such as LSTMs requires unfolding the network through time, introducing approximations that have not yet been fully characterised in the literature [47]. The implementation choices adopted for the CNN and CNN–BiLSTM models in this thesis are described in the Methodology chapter.

2.10 Perturbation-Based Evaluation of Attribution Methods

Attribution methods are only useful if their explanations faithfully reflect the model's actual reasoning. The literature distinguishes between interpretability, which describes how understandable an explanation is to a human, and faithfulness , which refers to how accurately an explanation reflects the model's internal computation [48]. An explanation

may appear visually convincing while failing to represent the features that genuinely influenced the model's prediction. Previous studies have shown that visually plausible attribution maps do not necessarily reflect the actual reasoning of a model, highlighting the need for quantitative evaluation in addition to visual inspection [49].

Perturbation-based evaluation provides an objective framework for assessing faithfulness without requiring human-annotated explanations. The underlying principle is that if an attribution method correctly identifies the most important input features, perturbing those features should cause the model's prediction to deteriorate more rapidly than perturbing less important features first [50]. In contrast, a gradual performance degradation suggests that the attribution method assigned a high importance to features that contributed relatively little to the prediction.

To systematically evaluate the quality of the attribution, the input features are perturbed according to their attributed importance. In the Most Relevant First (MoRF) setting, features with the highest attribution scores are perturbed first, whereas Least Relevant First (LeRF) perturbs features in the reverse order. These complementary perturbation orderings provide a general framework for assessing whether attribution scores are consistent with the behavior of the model [50]. Figure 13 illustrates this principle: under MoRF ordering, the target-class probability decreases rapidly because the most influential features are removed first, whereas under LeRF ordering, the prediction remains relatively stable until the least important features have been exhausted [51].

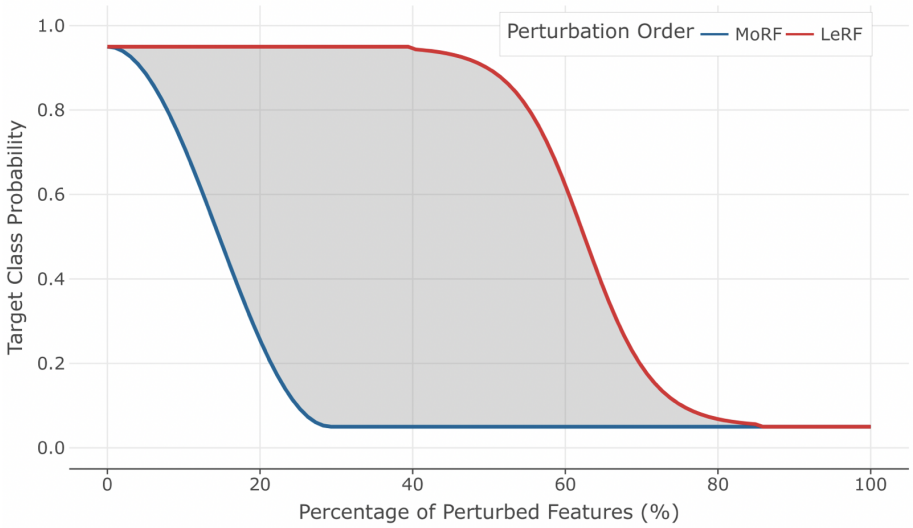


Figure 13: Illustration of perturbation curves under Most Relevant First (MoRF) and Least Relevant First (LeRF) perturbation. Under MoRF, perturbing highly relevant features causes a rapid decrease in the target-class probability, whereas LeRF perturbs less relevant features first, resulting in a slower degradation of the prediction. Adapted from [51].

Perturbation-based evaluation is commonly implemented using either zero-out masking,

which replaces selected features with zero, or interpolation, which substitutes them with reference values derived from a background distribution. These strategies provide complementary approaches for perturbing the input when evaluating attribution faithfulness and are widely adopted in the XAI literature [50, 52].

The resulting perturbation curve provides a quantitative description of how the model prediction changes as increasingly important features are perturbed. Within the perturbation-based evaluation framework, a faithful attribution method typically produces a steeper degradation curve under MoRF perturbation, indicating that highly ranked features have a greater influence on the model's prediction. To facilitate quantitative comparison across methods, perturbation curves are commonly summarised using the Area Under the Curve (AUC), estimated by the trapezoidal rule:

$$\mathrm{AUC} = \sum_{k=1}^{K-1} \frac{p_k + p_{k+1}}{2}(\alpha_{k+1} - \alpha_k), \tag{28}$$

where p_k denotes the model performance after perturbing a fraction $\alpha_k \in [0, 1]$ of the input features. Under MoRF perturbation, a lower AUC is generally interpreted as indicating higher attribution faithfulness, as the model performance deteriorates more rapidly when the most influential features are progressively perturbed [53].

Despite its widespread use, perturbation-based evaluation has several limitations. Perturbing the input may generate unrealistic or out-of-distribution samples that do not naturally occur in the original data. Consequently, the observed performance degradation may partially reflect the model's sensitivity to perturbed inputs rather than solely the importance of the perturbed features. In addition, the measured faithfulness can depend on the perturbation strategy and the evaluation metric employed. Perturbation-based metrics should therefore be regarded as useful proxies for attribution faithfulness rather than definitive measures of explanation quality [54].

2.11 Statistical Analysis for Comparing Attribution Methods

Comparing attribution methods quantitatively requires statistical tools that can handle paired, non-normally distributed evaluation metrics computed on the same set of ECG recordings. Two complementary procedures are used : the first, Wilcoxon signed-rank test, assess whether observed differences are statistically significant; and the second Bland–Altman analysis characterizes the magnitude and consistency of those differences.

2.11.1 Wilcoxon Signed-Rank Test

The Wilcoxon signed-rank test [55] is a non-parametric test for paired observations. Given n paired measurements (x_i, y_i), the signed differences $d_i = x_i - y_i$ are ranked by absolute value. Let R_i^+ and R_i^- denote the ranks of positive and negative differences, respectively. The test statistic is

$$W = \min\left(\sum_{d_i>0} R_i^+,\ \sum_{d_i<0} R_i^-\right). \tag{29}$$

Under the null hypothesis that the median difference is zero, W follows a known distribution that can be used to compute an exact or approximate p-value [56]. Pairs with $d_i = 0$ are excluded from the ranking and the ties are handled by assigning the average ranks.

2.11.2 Bland–Altman Analysis

Although the Wilcoxon test indicates whether a difference exists, it does not describe how large or consistent that difference is across the range of observed values. The Bland Altman analysis [57] addresses this by plotting, for each pair i, the mean per-pair $m_i = (x_i + y_i)/2$ on the horizontal axis against the difference $d_i = x_i - y_i$ on the vertical axis. Three horizontal reference lines summarize the plot: the mean bias

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \tag{30}$$

and the 95 % limits of agreement (LoA)

$$\bar{d} \pm 1.96\, s_d, \qquad s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2}, \tag{31}$$

which are expected to contain approximately 95 % of individual pairwise differences [57]. If the differences are randomly scattered around $\bar{d}$ without a systematic trend across m_i, the two methods agree consistently across the measurement range. Narrow limits of agreement indicate that the methods are practically interchangeable; wide limits indicate substantial divergence for individual observations, even if the mean bias is small.

3

Methodology

This chapter describes the methodology used to evaluate the faithfulness of explainable artificial intelligence (XAI) methods for deep learning models applied to ECG analysis. Although classification performance was initially evaluated on the five-class PTB-XL classification task, the explainability benchmark presented in this thesis focuses on the binary MI-versus-NORM classification task. This design enables the use of a physiologically meaningful healthy reference signal while simplifying the interpretation of attribution methods for myocardial infarction detection.

Two deep learning architectures, a convolutional neural network (CNN) and a CNN–BiLSTM with Attention, are evaluated using three attribution methods: Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD). The resulting attribution maps are converted into feature rankings that guide the perturbation-based evaluation.

The proposed benchmark evaluates attribution faithfulness by progressively perturbing the regions of the ECG signal ranked as most relevant by each attribution method and comparing the resulting prediction degradation with a matched random-order baseline. Perturbation is performed at both the segment and timestep levels using interpolation- and zero-out-based perturbation strategies.

Finally, attribution faithfulness is quantified using population-level perturbation curves and the corresponding area under the curve (AUC). Statistical significance is assessed using paired Wilcoxon signed-rank tests, while agreement between experimental conditions is evaluated using Bland–Altman analysis. The following sections describe each component of the proposed framework in detail.

Table 1: Overview of the three experimental stages of the proposed framework.

Stage	Purpose	Key Elements
Five-class Classification	Train and evaluate models on the full PTB-XL diagnostic taxonomy.	Five superclasses (NORM, MI, STTC, CD, HYP); CNN and CNN-BiLSTM-Attn architectures.
Binary MI–NORM Classification	Construct a focused dataset for explainability experiments.	Single-label recordings retained; NORM assigned label 0, MI label 1.
Attribution-guided Perturbation Benchmark	Assess faithfulness of attribution methods by measuring prediction degradation under guided perturbation.	IG, SmoothGrad-IG, DTD; zero-out and interpolation strategies; segment- and timestep-level granularity; random-order baseline; AUC, Wilcoxon test, Bland–Altman analysis.

3.1 Dataset and Preprocessing

3.1.1 PTB-XL Dataset

All experiments in this thesis are conducted using the PTB-XL dataset, a large publicly available 12-lead ECG database collected at the Physikalisch-Technische Bundesanstalt (PTB) and introduced by Wagner et al. [58]. The dataset contains 21,837 clinical 12-lead ECG recordings from 18,885 patients, each with a duration of ten seconds. Every recording is labelled with one or more SCP-ECG diagnostic codes, which are further grouped into five diagnostic superclasses: Normal sinus rhythm (NORM), Myocardial Infarction (MI), ST/T-change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). Recordings are available at two sampling rates; throughout this thesis, the 100 Hz version is used, resulting in signals of shape $(1000, 12)$, corresponding to 1,000 time samples for each of the 12 leads.

The data set provides a predefined stratified ten-fold partitioning variable (`strat_fold`) that ensures patient-level separation between data splits while maintaining balanced class distributions across folds [58]. This predefined partitioning is used without modification throughout all experiments.

3.2 Binary Task: NORM versus MI

For the explainability experiments, a binary dataset was created by selecting ECG recordings belonging to the Normal sinus rhythm (NORM) and Myocardial Infarction

(MI) diagnostic superclasses from the PTB-XL dataset. Focusing on these two classes enables a clinically meaningful evaluation of attribution methods while allowing the use of a healthy reference signal during perturbation.

To obtain unambiguous binary labels, each recording's SCP diagnostic codes were mapped to the NORM and MI diagnostic superclasses. Recordings associated with exactly one of these two target superclasses were retained, whereas recordings associated with both or neither target superclass were excluded. Additional diagnostic superclasses outside the binary target set were not used to determine the binary label. The retained recordings were encoded using binary labels, with NORM = 0 and MI = 1.

3.2.1 Dataset Split

The official patient-stratified ten-fold split provided with the PTB-XL dataset was adopted in this study. Folds 1–8 were used for training, fold 9 for validation, and fold 10 for testing. This split ensures that recordings from the same patient are not shared across different data partitions while maintaining a similar class distribution in each set.

3.2.2 Data Preprocessing

Each retained recording underwent the following sequence of preprocessing steps, applied in the order described, before being passed to the model

Band-pass filtering.
A fourth-order Butterworth band-pass filter with cut-off frequencies of 0.5 Hz and 40 Hz was applied to each recording. The lower cut-off removes low-frequency baseline wander arising from respiration and electrode movement, while the upper cut-off attenuates high-frequency components, including muscle artifacts that lie above the diagnostically relevant frequency range of the clinical ECG. The filter was implemented as a cascade of second-order sections (SOS) and applied using zero-phase forward–backward filtering to avoid phase distortion in the output signal. Filtering was performed along the temporal axis, treating each of the twelve leads independently.

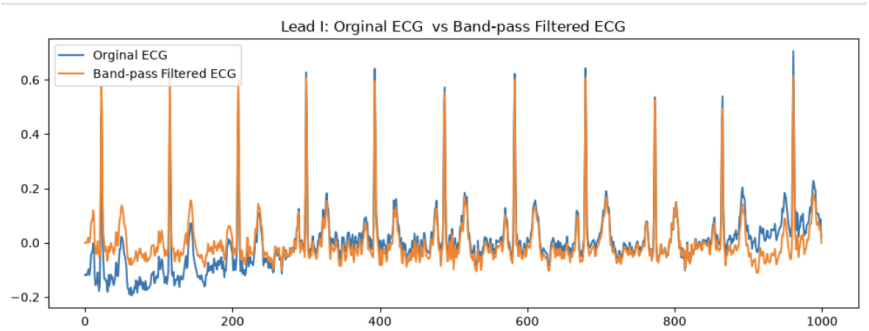


Figure 14: Band-pass filtering applied to Lead I of a representative PTB-XL recording. The original signal contains low-frequency baseline wander. The filtered signal retains the diagnostically relevant waveform morphology while suppressing drift below 0.5 Hz and noise above 40 Hz.

Per-lead z-score normalisation.
Following filtering, each lead was normalized to zero mean and unit variance. For a signal matrix of shape $(T, 12)$, the mean and standard deviation were computed independently over the $T = 1{,}000$ time samples of each lead, and the normalised value at time sample t for lead l is given by:

$$\tilde{x}_{t,l} = \frac{x_{t,l} - \mu_l}{\sigma_l + \varepsilon} \tag{32}$$

where μ_l and σ_l are the temporal mean and standard deviation of lead l, and $\varepsilon = 10^{-8}$ is a small constant included for numerical stability. This per-lead normalization places all leads on a comparable scale while reducing amplitude differences arising from electrode placement and inter-patient variability, ensuring consistent input scaling for the deep learning models.

Reshaping
After normalisation, each signal was transposed from the raw shape $(1000, 12)$ returned by the WFDB reader to the shape $(12, 1000)$ expected by the model architectures, placing the lead dimension first and the time dimension second. The final representation of each sample was therefore a two-dimensional array of shape $(12, 1{,}000)$ containing twelve normalised leads of 1,000 samples each.

The final preprocessed dataset consists of an input tensor $X \in \mathbb{R}^{N \times 12 \times 1000}$ and a corresponding label vector $y \in \{0, 1\}^N$, where N denotes the number of ECG recordings in the respective dataset split. The processed data were stored separately for the training, validation, and test sets for subsequent model training and evaluation.

3.3 Deep Learning Models

3.3.1 Convolutional Neural Network (CNN)

The first architecture which we evaluated in this study is a one-dimensional convolutional neural network designed for multichannel ECG classification. The model receives input tensors of shape $(B, 12, 1000)$, where B is the batch size, 12 is the number of ECG leads, and 1000 is the number of time samples per recording.

The feature extractor consists of three one-dimensional convolutional blocks with progressively increasing channel depths of 32, 64, and 128. The convolutional blocks use kernel sizes of 7, 5, and 3 with corresponding padding values of 3, 2, and 1, preserving the temporal resolution within each block before downsampling. Each block is followed by Batch Normalisation and a ReLU activation function. Max-pooling is applied after the first two blocks for temporal downsampling, while the final block is followed by adaptive average pooling, producing a 128-dimensional feature representation for each recording.

The classifier consists of a dropout layer with a rate of 0.2 followed by a fully connected layer that maps the learned representation to the output logits. The output dimension is $C = 5$ for the five-class classification task and $C = 2$ for the binary NORM-versus-MI classification task.

The model is trained using the AdamW optimiser with an initial learning rate of 10^{-3}, a weight decay of 10^{-4}, and a batch size of 128. The learning rate is reduced by a factor of 0.5 when the validation accuracy does not improve for three consecutive epochs, and gradient norms are clipped to 1.0 to improve training stability. The checkpoint achieving the highest validation accuracy is retained for all subsequent evaluation and explainability experiments.

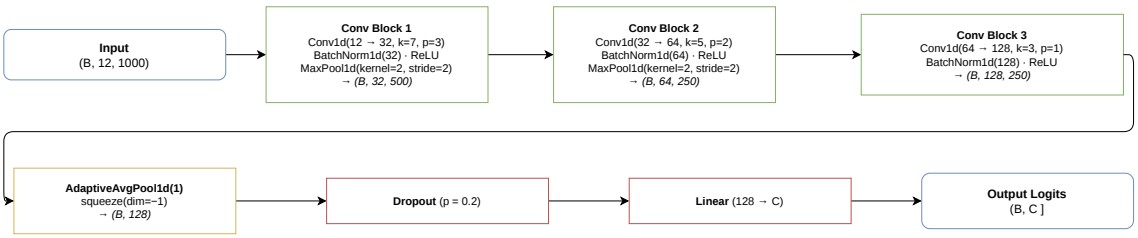


Figure 15: Architecture of the CNN used in this thesis. The final linear layer produces C output logits, where $C = 5$ for the five-class classification task and $C = 2$ for the binary NORM-versus-MI task.

3.3.2 Convolutional Neural Network–Bidirectional Long Short-Term Memory (CNN–BiLSTM) with Temporal Attention

The second architecture extends the CNN by combining the same one-dimensional convolutional feature extractor with a bidirectional Long Short-Term Memory (BiLSTM) network and a temporal attention mechanism. The convolutional stage is identical to that described for the CNN in Section 3.3.1, except that the final adaptive average pooling layer is omitted to preserve the temporal feature sequence for recurrent processing, resulting in an output tensor of shape $(B, 128, 250)$.

The feature tensor was transposed to $(B, 250, 128)$ so that each of the 250 time steps was represented by a 128-dimensional feature vector. The resulting sequence was processed by a single-layer bidirectional LSTM with a hidden size of 64 in each direction, producing a 128-dimensional output at every time step.

The temporal attention mechanism followed the formulation introduced in Section 2.5, with the BiLSTM hidden states used as the input sequence. In the implemented model, temporal aggregation was performed using a two-layer feed-forward attention network. For each time step, an attention score was computed, normalised using a softmax function over the temporal dimension, and used to obtain a weighted sum of the BiLSTM outputs:

$$e_t = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_t + \mathbf{b}), \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T} \exp(e_j)}, \qquad \mathbf{c} = \sum_{t=1}^{T'} \alpha_t \mathbf{h}_t, \tag{33}$$

where $\mathbf{h}_t \in \mathbb{R}^{128}$ denotes the BiLSTM output at time step t, e_t is the corresponding attention score, α_t is the normalised attention weight, and $\mathbf{c} \in \mathbb{R}^{128}$ is the resulting context vector.

A dropout layer with a rate of 0.3 was applied to the context vector before the final fully connected layer, which produced the output logits. The output dimension was $C = 5$ for the five-class classification task and $C = 2$ for the binary NORM-versus-MI classification task.

The CNN–BiLSTM model differed from the CNN only in architecture-specific settings, namely the dropout rate and the use of class weighting during training.

The two architectures were selected because they process temporal information in different ways. The CNN relies on convolutional operations with progressively increasing receptive fields and reduces the temporal feature sequence to a single representation using adaptive average pooling. In contrast, the CNN–BiLSTM with Attention preserves the temporal structure of the convolutional feature map, processes the resulting sequence using a bidirectional recurrent layer, and aggregates the BiLSTM outputs through learned attention weights.

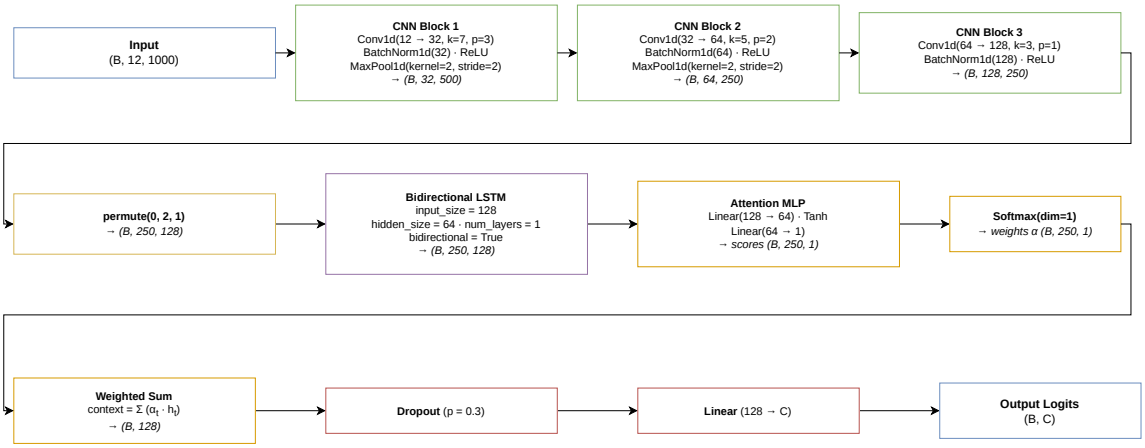


Figure 16: Architecture of the CNN–LSTM model with temporal attention. The final linear layer produces C output logits, where $C = 5$ for the five-class classification task and $C = 2$ for the binary NORM-versus-MI task.

Evaluating both architectures within the same attribution benchmark makes it possible to assess the faithfulness of the three attribution methods across models that differ in how temporal features are processed and aggregated. The implementation of the attribution methods for these architectures is described in the following section.

3.4 Classification Performance Metrics

The five-class classification models were evaluated using precision, recall, F1-score, macro F1-score, macro area under the receiver operating characteristic curve (Macro AUC), and macro average precision (Macro AP). Macro-averaged metrics assign equal weight to each diagnostic class and are therefore suitable for evaluating performance in the presence of class imbalance.

Precision. Precision measures the proportion of samples predicted as class c that are correctly assigned to that class. A high precision indicates that the model produces relatively few false-positive predictions.

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \tag{34}$$

where TP_c and FP_c denote the numbers of true-positive and false-positive predictions for class c, respectively.

Recall. Recall measures the proportion of samples belonging to class c that are correctly identified by the model. It reflects the model's ability to avoid false-negative predictions.

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \tag{35}$$

where FN_c denotes the number of false-negative predictions for class c.

F1-score. The F1-score is the harmonic mean of precision and recall. Provides a single measure that balances false-positive and false-negative errors.

$$F_{1,c} = 2\frac{\text{Precision}_c \, \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \tag{36}$$

where $F_{1,c}$ denotes the F1-score for class c.

Macro F1-score. The macro F1-score is obtained by averaging the class-specific F1-scores. Each class contributes equally to the final value, regardless of its number of samples.

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c}, \tag{37}$$

where $C = 5$ is the number of diagnostic classes.

Macro AUC. For each class, the receiver operating characteristic curve is computed using a one-versus-rest formulation. Its area measures how well the model separates that class from all remaining classes across classification thresholds.

$$\text{Macro AUC} = \frac{1}{C}\sum_{c=1}^{C} \text{AUC}_c, \tag{38}$$

where AUC_c is the one-versus-rest ROC AUC for class c.

Macro Average Precision. Average precision summarises the class-specific precision–recall curve and is particularly informative when class frequencies are unequal. Macro AP is obtained by assigning equal weight to the average precision of each class.

$$\text{Macro AP} = \frac{1}{C}\sum_{c=1}^{C} \text{AP}_c, \tag{39}$$

where AP_c denotes the one-versus-rest average precision for class c.

3.5 Integrated Gradients: Implementation Details

Integrated Gradients (IG) was used to generate class-specific attribution maps for each trained classification model. The implementation followed the formulation introduced in Section 2.7, while the following subsections describe the implementation choices used in this study.

Let the preprocessed ECG recording be denoted by $x \in \mathbb{R}^{L\times T}$, where $L = 12$ is the number of ECG leads and $T = 1000$ is the number of time samples. Let x' denote the reference baseline and F the trained classification model. For a target class c, Integrated Gradients was computed with respect to the corresponding pre-softmax logit $F_c(x)$. In the perturbation benchmark, the target class corresponded to the class of the evaluated sample, namely $c = 0$ for NORM and $c = 1$ for MI. The resulting attribution map $A \in \mathbb{R}^{L\times T}$ assigns an attribution value to each lead–time position of the input signal.

Following the formulation introduced in Chapter 2, the attribution assigned to feature i is

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F_c(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha. \tag{40}$$

For IG, the completeness property was checked by comparing the sum of all feature attributions with the difference between the target-class logits obtained for the input and the corresponding baseline.

The path integral was approximated using the midpoint Riemann rule with $m = 50$ uniformly spaced interpolation steps. The interpolation coefficient for the k-th step was defined as

$$\alpha_k = \frac{k - \frac{1}{2}}{m}, \qquad k = 1, \ldots, m, \tag{41}$$

where $m = 50$. The gradients evaluated along the interpolation path were averaged to approximate the integral.

At each interpolation point, the gradient of the target-class logit with respect to the interpolated input was computed using automatic differentiation. The average gradient was then multiplied element-wise by $(x - x')$ to obtain the Integrated Gradients attribution map. A derivation of the Integrated Gradients completeness property is provided in Appendix E.

To assess the numerical approximation, the completeness property was evaluated for each Integrated Gradients attribution map by comparing the sum of all attribution values with the difference between the target-class logits obtained for the input and the corresponding baseline:

$$\Delta_{\text{comp}} = \frac{\left|\sum_i \widehat{IG}_i(x) - (F_c(x) - F_c(x'))\right|}{|F_c(x) - F_c(x')| + \varepsilon}, \tag{42}$$

Here, $\hat{\text{IG}}(x)$ denotes the m-step Riemann approximation of $\text{IG}(x)$, and $\varepsilon = 10^{-8}$ prevents division by zero. Since the path integral was approximated numerically using a finite number of interpolation steps, a small discrepancy was expected. The completeness error was therefore used as a diagnostic measure of numerical accuracy rather than as a strict pass/fail criterion.

To further validate the Integrated Gradients implementation, we evaluated the completeness property on 20 MI test samples for both architectures as shown in Table 2. The CNN achieved a mean relative completeness error of 8.51×10^{-3}, while the CNN–BiLSTM with Attention achieved 1.77×10^{-2} as we can see in table 2 . These low errors indicate good agreement with the theoretical completeness property and provide additional confidence in the numerical correctness of the Integrated Gradients implementation

Model	**Mean Rel. Error**	**Median Rel. Error**
CNN	8.51×10^{-3}	5.19×10^{-3}
CNN–BiLSTM with Attention	1.77×10^{-2}	8.01×10^{-3}

Table 2: Integrated Gradients completeness validation for the two evaluated architectures. The completeness property was assessed on 20 MI test samples.

The final output of Integrated Gradients is a signed attribution map $A \in \mathbb{R}^{L \times T}$ with the same dimensions as the input ECG signal. Positive attribution values indicate a positive contribution to the target-class logit relative to the baseline, whereas negative values indicate a negative contribution. The absolute attribution magnitude was subsequently used to rank signal regions for the perturbation experiments described in Section 3.8. Throughout attribution generation, model parameters remained fixed, and gradients were computed only with respect to the input.

Algorithm 1 Integrated Gradients Attribution

Require Trained model F; input $x \in \mathbb{R}^{L\times T}$; baseline $x' \in \mathbb{R}^{L\times T}$; target class c; number of integration steps $m = 50$

Ensure: Attribution map $A \in \mathbb{R}^{L\times T}$; completeness diagnostic Δ_{comp}

$G \leftarrow 0^{L\times T}$

for $k = 1, \ldots, m$ **do**

  $\alpha_k \leftarrow \dfrac{k - \frac{1}{2}}{m}$ ▷ Midpoint of the k-th integration interval

  $z_k \leftarrow x' + \alpha_k(x - x')$ ▷ Construct the interpolated input

  Enable gradient tracking for the interpolated input z_k

  $s_k \leftarrow F_c(z_k)$ ▷ Compute the target pre-softmax logit

  $g_k \leftarrow \dfrac{\partial F_c(z_k)}{\partial z_k}$ ▷ Compute the gradient with respect to the interpolated input

  $G \leftarrow G + g_k$ ▷ Accumulate the gradient

end for

$\overline{G} \leftarrow \dfrac{G}{m}$ ▷ Compute the mean gradient over all midpoint samples

$A \leftarrow (x - x') \odot \overline{G}$ ▷ Compute the attribution map

$\Delta_{\text{comp}} \leftarrow \dfrac{|\sum_i A_i - (F_c(x) - F_c(x'))|}{|F_c(x) - F_c(x')| + \varepsilon}$ ▷ Compute the relative completeness diagnostic

Return A, Δ_{comp}



Figure 17: Integrated Gradients attribution map for a correctly classified MI sample (Lead II) shown as a representative example produced by the CNN-BiLSTM-Attn model . Warmer colours indicate larger absolute Integrated Gradients attribution magnitudes.

3.6 SmoothGrad-IG: Implementation Details

SmoothGrad-IG was used to reduce local variability in the attribution maps by averaging Integrated Gradients attributions computed from multiple noisy versions of the same ECG recording. For each noisy realisation, Integrated Gradients was computed using the procedure described in Section 2.8.

Let $x \in \mathbb{R}^{L\times T}$ denote the input ECG, x' the reference baseline, and F the trained classification model. For each of n independently sampled Gaussian noise realisations, a noisy input was generated as

$$\tilde{x}^{(j)} = x + \epsilon^{(j)}, \qquad \epsilon^{(j)} \sim \mathcal{N}(0, \sigma^2 I), \qquad j = 1, \ldots, n. \tag{43}$$

Integrated Gradients was then computed independently for each noisy input using the same baseline, target class, and numerical integration procedure:

$$A^{(j)} = IG\Big(\tilde{x}^{(j)}; x', c\Big)\,. \tag{44}$$

The final SmoothGrad-IG attribution map was obtained by averaging the attribution maps element-wise:

$$A^{\mathrm{SG}} = \frac{1}{n} \sum_{j=1}^{n} A^{(j)}. \tag{45}$$

The implementation used $n = 5$ noisy realisations. The Gaussian noise standard deviation was set to $\sigma = 0.05$ on the z-score-normalised input scale. For each noisy input, Integrated Gradients was computed using the same reference baseline and target class, together with $m = 50$ midpoint integration steps. After all attribution maps had been computed, they were averaged element-wise to obtain the final SmoothGrad-IG attribution map. The resulting map had the same dimensions as the original ECG signal and was subsequently used in the perturbation benchmark described in Section 3.8.

Algorithm 2 SmoothGrad-Integrated Gradients Attribution

Require: Trained model F; input x; baseline x'; target class c; number of noisy realisations $n = 5$; integration steps $m = 50$; Gaussian noise standard deviation $\sigma = 0.05$
Ensure: SmoothGrad-IG attribution map A^{SG}
Initialise an empty list $\mathcal{A}$
for $j = 1, \ldots, n$ **do**
    Sample Gaussian noise $\epsilon^{(j)} \sim \mathcal{N}(0, \sigma^2 I)$
    $\tilde{x}^{(j)} \leftarrow x + \epsilon^{(j)}$
    $A^{(j)} \leftarrow \textsc{IntegratedGradients}(F, \tilde{x}^{(j)}, x', c, m)$
    Append $A^{(j)}$ to $\mathcal{A}$
end for
Compute the element-wise mean

$$A^{SG} = \frac{1}{n} \sum_{j=1}^{n} A^{(j)}.$$

Return A^{SG}



Figure 18: SmoothGrad-IG attribution map for a correctly classified MI sample (Lead II), produced by the CNN-BiLSTM-Attn model. Warmer colours indicate larger absolute Integrated Gradients attribution magnitudes.

3.7 Deep Taylor Decomposition: Implementation Details

We implement Deep Taylor Decomposition (DTD) using the z^+ redistribution rule to generate class-specific relevance maps for the trained models. Since the CNN and the CNN–BiLSTM with Attention differ substantially in architecture, separate propagation procedures were used. Full layer-wise z^+ propagation was applied to the CNN, whereas a hybrid implementation was required for the CNN–BiLSTM with Attention.

Let $F(x)$ denote the vector of pre-softmax logits produced by the trained model for an input $x \in \mathbb{R}^{L\times T}$, and let c denote the target class. For the CNN, relevance propagation was initialised at the output layer by assigning relevance only to the target-class logit:

$$R_j^{(L)} = \begin{cases} \max(F_c(x), \varepsilon), & j = c, \\ 0, & j \neq c, \end{cases} \tag{46}$$

where $\varepsilon = 10^{-9}$ was used for numerical stabilisation.

For linear and convolutional layers, relevance was propagated using the z^+ redistribution rule. Let

$$a_i^+ = \max(a_i, 0), \qquad w_{ij}^+ = \max(w_{ij}, 0), \qquad z_{ij}^+ = a_i^+ w_{ij}^+. \tag{47}$$

The relevance assigned to unit i in layer l was then computed as

$$R_i^{(l)} = \sum_j \frac{z_{ij}^+}{\sum_k z_{ik}^+ + \varepsilon} R_j^{(l+1)}. \tag{48}$$

The CNN and the CNN–BiLSTM with Attention required different propagation procedures because of their architectural differences.

For the CNN, relevance was propagated backwards through the fully connected classifier, adaptive average pooling layer, convolutional blocks, and max-pooling layers. The dropout layer was inactive in evaluation mode and was therefore treated as a pass-through operation. Linear and convolutional layers used the z^+ redistribution rule. Max-pooling relevance was routed back to the winning input positions using the pooling indices recorded during the forward pass. Batch Normalisation and ReLU layers were treated as pass-through operations and did not explicitly redistribute relevance.

For the CNN–BiLSTM with Attention, exact layer-wise DTD propagation was not applied through the recurrent and attention components. Instead, the gradient of the target-class

logit was backpropagated through the classifier, attention mechanism, and BiLSTM to the CNN feature representation H. Relevance at this interface was approximated using

$$R_{\text{interface}} = \left(\frac{\partial F_c(x)}{\partial H} \odot H \right)_+ , \tag{49}$$

where $(\cdot)_+$ denotes retention of positive contributions only. The resulting interface relevance was then propagated through the convolutional feature extractor using the same z^+ redistribution procedure as for the CNN.

In both implementations, a small constant ε was added to the denominators for numerical stability, and activations and weights were restricted to their positive components during z^+ propagation. Relevance was propagated backwards until a relevance value was assigned to every lead–time position of the input ECG.

The final output of Deep Taylor Decomposition was a non-negative relevance map $R \in \mathbb{R}^{L \times T}$ with the same dimensions as the input ECG signal. Each value represents the relevance assigned to the corresponding lead–time position for the target-class prediction. Larger relevance values indicate that more relevance was assigned to that position, whereas values close to zero indicate little assigned relevance. The resulting relevance map was subsequently used to rank signal regions in the perturbation benchmark described in Section 3.8.

Algorithm 3 DTD Attribution for the CNN

Require: Trained CNN model F; input ECG $x \in \mathbb{R}^{L \times T}$; target class c; stabiliser $\varepsilon = 10^{-9}$
Ensure: Input relevance map $R^{(0)} \in \mathbb{R}^{L \times T}$

1: Perform a forward pass through $F(x)$ and store the input activations of all eligible layers.
2: Initialise the output relevance:

$$R_j^{(L)} = \begin{cases} \max(F_c(x), \varepsilon), & j = c, \\ 0, & j \neq c. \end{cases}$$

3: Propagate relevance through the classifier using the z^+ redistribution rule, treating dropout as a pass-through operation.
4: Redistribute relevance through the adaptive average pooling layer in proportion to the positive pre-pooling activations.
5: **for** each convolutional block in reverse order **do**
6: Apply the z^+ redistribution rule to the convolutional layer
7: Invert the corresponding max-pooling layer using pooling indices recomputed from the stored layer input, when present.
8: **end for**
9: Treat ReLU and Batch Normalisation layers as pass-through operations throughout relevance propagation.
10: Obtain the input relevance map $R^{(0)}$.
11: Return $R^{(0)}$

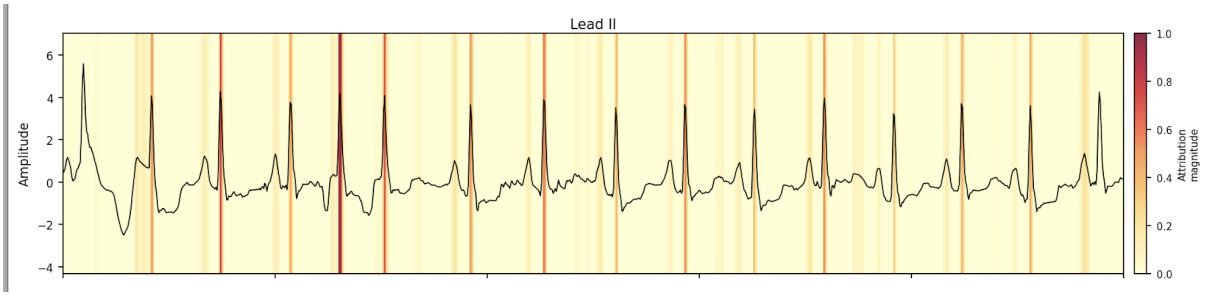


Figure 19: DTD relevance map for a correctly classified MI sample (Lead II), produced by the CNN model. Warmer colours indicate higher relevance magnitude assigned by the z^+ redistribution rule.

Algorithm 4 Hybrid DTD attribution for the CNN–BiLSTM with Attention

Require: Trained CNN–BiLSTM with Attention model F; input ECG $x \in \mathbb{R}^{L \times T}$; target class c; stabiliser $\varepsilon = 10^{-9}$

Ensure: Input relevance map $R^{(0)} \in \mathbb{R}^{L \times T}$

1: Perform a forward pass through the CNN feature extractor and store the input activations of all eligible layers.

2: Enable gradient computation for the CNN feature representation H.

3: Forward H through the bidirectional LSTM, temporal attention mechanism, and classifier to obtain the target-class logit $F_c(x)$.

4: Compute the gradient of the target-class logit with respect to the CNN feature representation:

$$G_H \leftarrow \frac{\partial F_c(x)}{\partial H}.$$

5: Form the CNN–BiLSTM interface relevance by multiplying the CNN feature activations element-wise by their corresponding gradients and retaining only positive contributions:

$$R_{\text{interface}} \leftarrow (G_H \odot H)_+ .$$

6: **for** each convolutional block in reverse order **do**

7: Apply the z^+ redistribution rule to the convolutional layer.

8: Recompute the corresponding max-pooling indices from the stored layer input and invert the pooling operation using max-unpooling, when present.

9: Treat ReLU and Batch Normalisation layers as pass-through operations.

10: **end for**

11: Obtain the input relevance map $R^{(0)}$.

12: **return** $R^{(0)}$

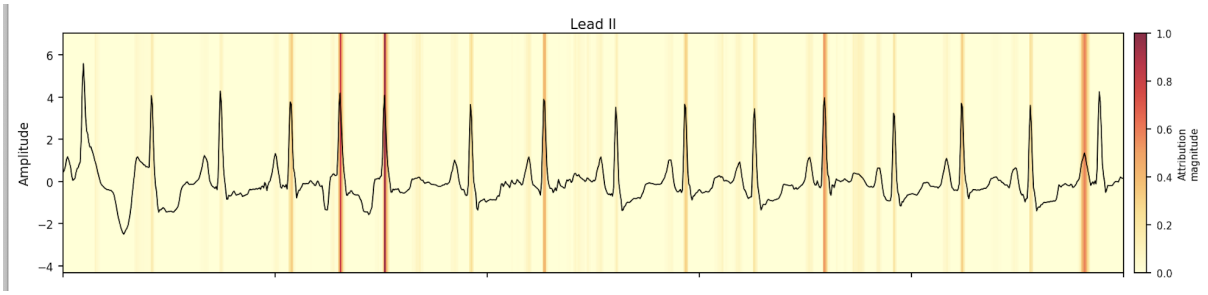


Figure 20: DTD relevance map for a correctly classified MI sample (Lead II), produced by the CNN-LSTM-Attn model using the hybrid z^+ procedure. Warmer colours indicate higher relevance magnitude.

The resulting attribution maps from all three methods are subsequently converted into ranked temporal importance orderings, a procedure described in the following section.

3.8 Attribution-guided Perturbation Benchmark

The perturbation benchmark evaluates the faithfulness of each attribution method by measuring how model performance changes when the input ECG is progressively modified according to the attribution ranking. Each attribution map is converted into an ordered sequence of temporal perturbation units, consisting of either contiguous segments or ranked timesteps. For the segment-level analysis, segments are perturbed sequentially, whereas the timestep-level analysis perturbs groups of ten ranked timesteps at each step.

The model is evaluated after each cumulative perturbation step, producing a prediction trajectory over the full perturbation range. The same procedure is repeated using a random ordering of the corresponding perturbation units, yielding a matched baseline trajectory. Comparing the attribution-guided and random-order trajectories assesses whether perturbing the regions prioritised by an attribution method has a greater effect on model performance than perturbing randomly selected regions.

3.8.1 Segment-level Attribution-guided Perturbation

Edge masking. Before ranking, the attribution map $A \in \mathbb{R}^{L\times T}$ was edge-masked to reduce potential boundary effects. The first and last $M = 30$ timestep columns were set to zero:

$$\tilde{A}_{l,t} = \begin{cases} 0 & t < M \text{ or } t \geq T - M, \\ A_{l,t} & \text{otherwise.} \end{cases} \tag{50}$$

All subsequent importance scoring and ranking were performed using $\tilde{A}$.

Segment partitioning. The temporal axis was divided into $N_{\text{seg}} = 50$ contiguous segments. Each segment had length

$$q = \left\lfloor \frac{T}{N_{\text{seg}}} \right\rfloor = 20 \tag{51}$$

timesteps, with the final segment extended to the end of the recording. For $k = 0, \ldots, N_{\text{seg}} - 1$, the index set of segment k was defined as

$$\mathcal{I}_k = \begin{cases} \{kq,\ kq+1,\ \ldots,\ (k+1)q-1\}, & k < N_{\text{seg}} - 1, \\ \{kq,\ kq+1,\ \ldots,\ T-1\}, & k = N_{\text{seg}} - 1. \end{cases} \tag{52}$$

For $T = 1{,}000$ and $N_{\text{seg}} = 50$, all segments contained exactly 20 timesteps. The definition above also accommodates cases in which T is not exactly divisible by N_{seg}.

Segment importance scoring. The importance score of segment k was computed as the sum of the absolute attribution values across all $L = 12$ leads and all timesteps belonging to that segment:

$$S_k = \sum_{l=1}^{L} \sum_{t \in \mathcal{I}_k} \left| \tilde{A}_{l,t} \right|, \qquad k = 0, 1, \ldots, N_{\text{seg}} - 1. \tag{53}$$

The segments were ranked in descending order of S_k:

$$\pi = \left(u_1, u_2, \ldots, u_{N_{\text{seg}}} \right) = \text{argsort}_k(-S_k)\,, \tag{54}$$

where u_1 denotes the highest-ranked segment and $u_{N_{\text{seg}}}$ the lowest-ranked segment.

Timestep importance scoring. At the finer temporal granularity, an importance score was assigned to each timestep by averaging the absolute attribution values across all L leads:

$$a_t = \frac{1}{L} \sum_{l=1}^{L} \left| \tilde{A}_{l,t} \right|, \qquad t = 0, 1, \ldots, T - 1. \tag{55}$$

The T timestep indices were then ranked in descending order of a_t. During the perturbation analysis, this ordering determined which groups of ten timesteps were modified at each cumulative perturbation step.

3.8.2 Synthetic Healthy ECG Baseline

For MI recordings, a heart-rate-matched synthetic healthy ECG was generated as the reference signal for interpolation-based perturbation. The reference is denoted by $x_{\text{ref}} \in \mathbb{R}^{L \times T}$. Its construction proceeded as follows.

Heart-rate estimation.
R-peaks were detected in Lead II of the original MI recording, and the heart rate was estimated from the mean R–R interval as

$$\mathrm{HR} = \frac{60}{\Delta t_{\mathrm{RR}}} \quad [\mathrm{bpm}], \tag{56}$$

where Δt_{RR} denotes the mean interval between consecutive R-peaks in seconds. If fewer than two reliable R-peaks were detected, a default heart rate of 75 bpm was used. Before signal generation, the estimated heart rate was restricted to the range 45–140 bpm.

ECG simulation and normalisation.
A synthetic healthy ECG was generated at the estimated heart rate using the NeuroKit2 `ecg_simulate` function. The `ecgsyn` simulation method was used whenever possible, with the `simple` method serving as a fallback. The generated one-dimensional signal was extended to 12 leads by applying an independently sampled circular temporal shift between -5 and $+5$ samples to each lead. Each synthetic lead was then independently normalised to zero mean and unit variance, consistent with the scale of the preprocessed input ECGs.

R-peak alignment.
The normalised synthetic ECG was temporally aligned with the original MI recording by matching their central R-peaks. The alignment shift was defined as

$$\Delta t = r_{\mathrm{MI}} - r_{\mathrm{syn}}, \tag{57}$$

where r_{MI} and r_{syn} denote the central R-peak sample indices of the MI recording and synthetic ECG, respectively. The common shift was applied to all synthetic leads using boundary-padded translation without circular wrapping.

For NORM recordings, no synthetic reference was generated. Instead, $x_{\mathrm{ref}} = \mathbf{0}$ was used, and perturbation was restricted to zero-out masking. Interpolation toward a synthetic healthy ECG was not applied to NORM recordings.

3.9 Perturbation Strategies

Two perturbation strategies are used in the reported experiments.

Zero-out. For zero-out perturbation, each selected temporal position is replaced with the constant value zero:

$$x_{\mathrm{cur},l,t} \leftarrow 0, \qquad l = 1, \ldots, L,\ t \in \mathcal{J}, \tag{58}$$

where $\mathcal{J}$ denotes the set of perturbed temporal positions. All values within the selected region are replaced by zero while the remaining parts of the ECG signal remain unchanged.

Interpolation. Each selected position is replaced by a fixed convex mixture of the original MI signal and the aligned synthetic reference:

$$x_{\mathrm{cur},l,t} \leftarrow (1-\alpha)\, x_{\mathrm{MI},l,t} + \alpha\, x_{\mathrm{ref},l,t}, \qquad \alpha = 0.5 \tag{59}$$

for all l, $t \in \mathcal{J}$. The blend is always computed from the *original* MI signal values $x_{\mathrm{MI},l,t}$, not from the progressively modified working copy x_{cur}. The replacement value for each region is therefore determined once by the original recording and the reference, and is not affected by prior perturbation steps; however, the replacement is applied cumulatively to x_{cur}, so previously replaced regions retain their modified values.

3.10 Segment-level Perturbation

Given the ranked sequence π, the benchmark constructs a prediction trajectory by replacing one additional segment at each step following the attribution-guided order, from the most to the least important. Replacements are cumulative: after step m, all segments $u_1, \ldots, u_m$ have been replaced in x_{cur}, whereas the remaining $N_{\mathrm{seg}} - m$ segments retain their original values. The model is evaluated after each replacement, and the corresponding population-level soft balanced accuracy is recorded. The complete trajectory contains $N_{\mathrm{seg}} + 1 = 51$ points, with step 0 corresponding to the unmodified signal. Algorithms 5 and 6 describe the zero-out and interpolation perturbation strategies, respectively.

Figures 21 and 22 present representative examples of the interpolation and zero-out perturbation strategies used in this work. The highlighted regions correspond to the highest-ranked segments identified by the attribution method and are progressively replaced according to the selected perturbation strategy.

Algorithm 5 Segment-level Attribution-guided Zero-out Perturbation

Require: Trained model F; ECG $x \in \mathbb{R}^{L\times T}$; attribution map $A \in \mathbb{R}^{L\times T}$; target class c; precomputed cross-class probability trajectory p_{cross}; number of segments $N_{\text{seg}} = 50$; edge mask $M = 30$

Ensure: Soft balanced-accuracy trajectory $\mathbf{B} = (B_0, B_1, \ldots, B_{N_{\text{seg}}})$

Apply edge masking to the attribution map.

Partition the ECG into N_{seg} contiguous segments I_k and compute their importance scores.

Rank the segments in descending order of importance.

Initialise the working copy:

$$x_{\text{cur}} \leftarrow x.$$

Compute the unperturbed soft balanced accuracy:

$$p_{\text{this},0} = \text{softmax}(F(x_{\text{cur}}))_c, \qquad B_0 = \frac{p_{\text{this},0} + p_{\text{cross},0}}{2}.$$

for $m = 1, \ldots, N_{\text{seg}}$ **do**

    Select the next highest-ranked segment I_{u_m}.

    Replace the selected segment with zero:

$$x_{\text{cur}}[:, I_{u_m}] \leftarrow 0.$$

    Compute the updated soft balanced accuracy.

end for

return $\mathbf{B} = (B_0, B_1, \ldots, B_{N_{\text{seg}}})$.

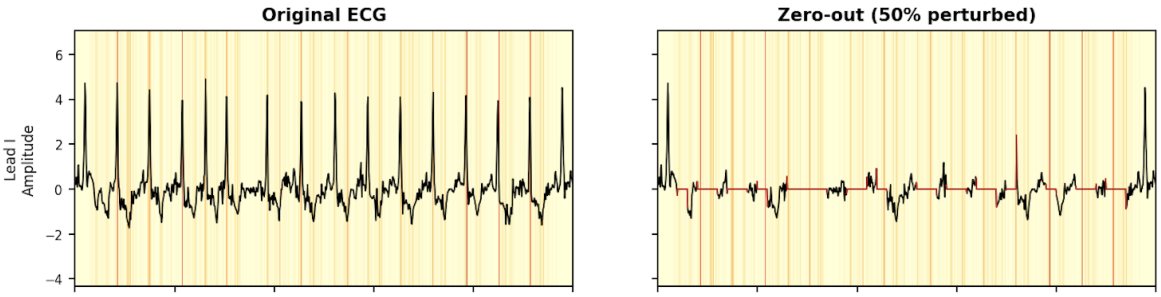


Figure 21: Example ECG before (left) and after (right) segment-level zero-out perturbation, where 50% of the highest-ranked segments (identified by the attribution method) have been replaced by the zero baseline. Red shading indicates highly attributed regions.

Algorithm 6 Segment-level Attribution-guided Interpolation Perturbation

Require: Trained model F; ECG $x \in \mathbb{R}^{L\times T}$; attribution map $A \in \mathbb{R}^{L\times T}$; aligned synthetic reference $x_{\text{ref}} \in \mathbb{R}^{L\times T}$; target class c; precomputed cross-class probability trajectory $\mathbf{p}_{\text{cross}}$; $\alpha = 0.5$; $N_{\text{seg}} = 50$; edge mask $M = 30$

Ensure: Soft balanced-accuracy trajectory $\mathbf{B} = (B_0, B_1, \ldots, B_{N_{\text{seg}}})$

Apply edge masking to the attribution map:

Partition the ECG into N_{seg} contiguous segments I_k and compute their importance scores

Rank the segments:

Initialise

$$x_{\text{cur}} \leftarrow x.$$

Compute the unperturbed soft balanced accuracy:

$$p_{\text{this},0} = \text{softmax}(F(x_{\text{cur}}))_c, \qquad B_0 = \frac{p_{\text{this},0} + p_{\text{cross},0}}{2}.$$

for $m = 1, \ldots, N_{\text{seg}}$ **do**

    Select segment I_{u_m}.

    Replace the selected segment using interpolation

$$x_{\text{cur}}[:, I_{u_m}] \leftarrow (1-\alpha)x[:, I_{u_m}] + \alpha x_{\text{ref}}[:, I_{u_m}],$$

    using the original ECG x rather than the progressively modified signal.

    Compute the updated soft balanced accuracy

end for

return $\mathbf{B} = (B_0, \ldots, B_{N_{\text{seg}}})$

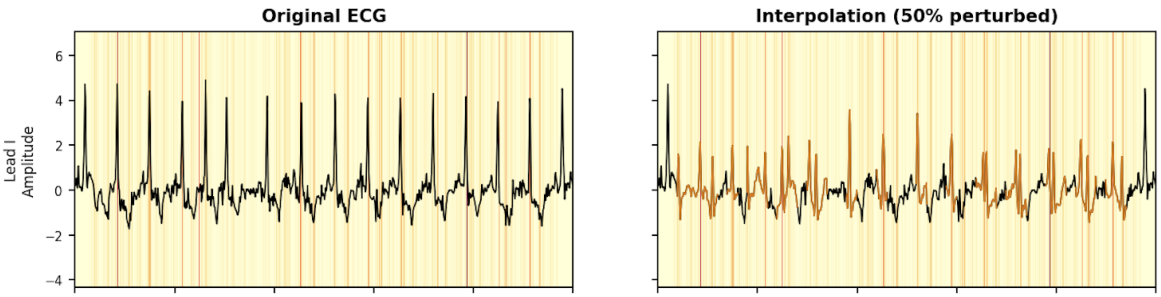


Figure 22: Example ECG before (left) and after (right) segment-level interpolation perturbation, where 50% of the highest-ranked segments have been replaced using interpolation with the aligned synthetic healthy reference ECG. Red shading indicates highly attributed regions.

3.11 Timestep-level Perturbation

At the finer granularity, all $T = 1{,}000$ timestep indices are ranked according to their importance score a_t (Equation 55) and perturbed in groups of $C = 10$ timesteps per step. Perturbing one timestep at a time would require 1,000 perturbation steps per sample. Grouping timesteps into chunks of $C = 10$ reduces this to $\lfloor T/C \rfloor = 100$ perturbation steps, resulting in a trajectory of $\lfloor T/C \rfloor + 1 = 101$ evaluations, including the unperturbed signal.

The indices within each group are not required to be contiguous in time. Instead, each group contains the next C highest-ranked timesteps in the attribution ordering, which may be located anywhere in the recording. Consequently, two adjacent entries in π^{ts} may correspond to widely separated positions if the attribution map assigns high importance to non-contiguous regions.

The perturbation rules are identical to those defined in Section 3.9, with the temporal index set $\mathcal{J} = \mathcal{G}_m$. Algorithms 7 and 8 describe the zero-out and interpolation variants, respectively. As the primary evaluation in this thesis is based on segment-level perturbation, the timestep-level perturbation results are provided in Appendix.

Algorithm 7 Timestep-level Attribution-guided Zero-out Perturbation

Require: Trained model F; ECG $x \in \mathbb{R}^{L\times T}$; attribution map $A \in \mathbb{R}^{L\times T}$; target class c; precomputed cross-class probability trajectory $\mathbf{p}_{\text{cross}}$; chunk size $C = 10$; edge mask $M = 30$

Ensure: Soft balanced-accuracy trajectory $\mathbf{B} = (B_0, B_1, \ldots, B_{\lfloor T/C \rfloor})$

1: Apply edge masking to A.

2: Compute timestep importance

$$a_t = \frac{1}{L}\sum_{l=1}^{L} |\widetilde{A}_{l,t}|.$$

3: Rank all timesteps:

$$\pi^{ts} = \text{argsort}(-a).$$

4: Initialise

$$x_{\text{cur}} \leftarrow x.$$

5: Compute the unperturbed soft balanced accuracy:

$$p_{\text{this},0} = \text{softmax}(F(x_{\text{cur}}))_c, \qquad B_0 = \frac{p_{\text{this},0} + p_{\text{cross},0}}{2}.$$

6: **for** $m = 1, \ldots, \lfloor T/C \rfloor$ **do**

7: Select the next group

$$G_m = \{\pi^{ts}[(m-1)C], \ldots, \pi^{ts}[mC-1]\}.$$

8: Replace the selected timesteps by zero-out:

$$x_{\text{cur}}[:, G_m] \leftarrow 0.$$

9: Compute the updated soft balanced accuracy.

10: **end for**

11: **return** $\mathbf{B} = (B_0, B_1, \ldots, B_{\lfloor T/C \rfloor})$

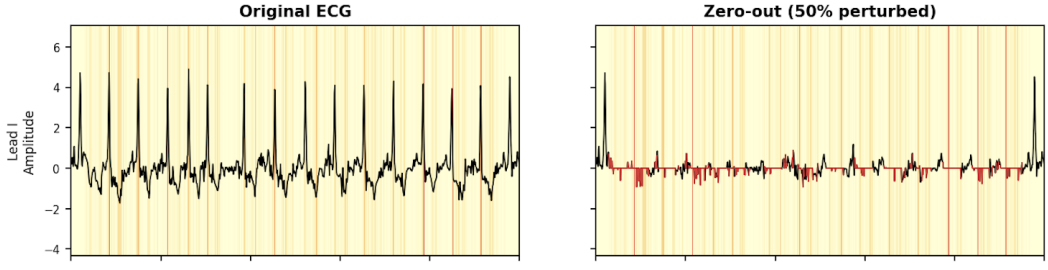


Figure 23: Example ECG before (left) and after (right) time step zero-out perturbation, where 50% of the highest-ranked timesteps (identified by the attribution method) have been replaced by the zeros . Red shading indicates highly attributed regions.

Algorithm 8 Timestep-level Attribution-guided Interpolation Perturbation

Require: Trained model F; ECG $x \in \mathbb{R}^{L\times T}$; attribution map $A \in \mathbb{R}^{L\times T}$; reference signal $x_{\text{ref}} \in \mathbb{R}^{L\times T}$; $\alpha = 0.5$; $C = 10$; $M = 30$; population cross-class probabilities $\left(p_{\text{cross},0}, \ldots, p_{\text{cross},\lfloor T/C \rfloor}\right)$

Ensure: Soft balanced-accuracy trajectory $B = (B_0, B_1, \ldots, B_{\lfloor T/C \rfloor})$

1: Apply edge masking to A.

2: Compute timestep scores

$$a_t = \tfrac{1}{L}\sum_{l=1}^{L} |\tilde{A}_{l,t}|.$$

3: Rank all timesteps:

$$\pi^{ts} = \text{argsort}(-a).$$

4: Initialise the working copy:

$$x_{\text{cur}} \leftarrow x.$$

5: Compute the unperturbed soft balanced accuracy:

$$p_{\text{this},0} = \text{softmax}(F(x_{\text{cur}}))_c, \qquad B_0 = \frac{p_{\text{this},0} + p_{\text{cross},0}}{2}.$$

6: **for** $m = 1, \ldots, \lfloor T/C \rfloor$ **do**

7: Select the next timestep group

$$G_m = \{\pi^{\text{ts}}[(m-1)C], \ldots, \pi^{\text{ts}}[mC-1]\}.$$

8: Replace the selected timesteps using interpolation from the original ECG:

$$x_{\text{cur}}[:, G_m] \leftarrow (1-\alpha)x[:, G_m] + \alpha x_{\text{ref}}[:, G_m].$$

9: Compute the updated soft balanced accuracy.

10: **end for**

11: **return B** $= (B_0, \ldots, B_{\lfloor T/C \rfloor})$

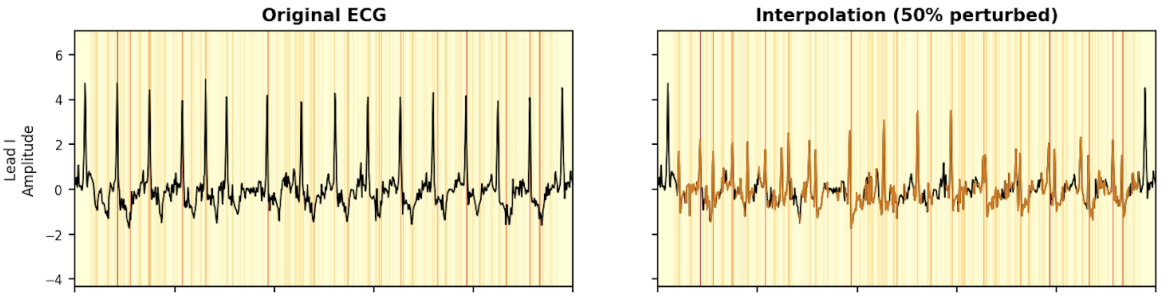


Figure 24: Example ECG before (left) and after (right) timesteps-level interpolation perturbation, where 50% of the highest-ranked timesteps have been replaced using interpolation with values from the aligned synthetic healthy reference ECG. Red shading indicates highly attributed regions.

3.12 Evaluation Metrics

Soft balanced accuracy. Because each test sample belongs to exactly one class, a single perturbation trajectory provides only one component of balanced accuracy. To obtain a per-sample metric, the model probability assigned to the sample's true class is combined with a precomputed population-level probability of the opposite class.

For a test sample at perturbation step k, the probability assigned to the sample's true class is

$$p_k^{(\text{same})} = P\Big(y_{\text{true}} \mid x_{\text{cur}}^{(k)}\Big)\,. \tag{60}$$

The per-sample soft balanced accuracy is then defined as

$$\text{BA}(k) = \frac{1}{2}\left(p_k^{(\text{same})} + p_k^{(\text{cross})}\right), \tag{61}$$

where $p_k^{(\text{same})}$ denotes the model probability assigned to the sample's true class and $p_k^{(\text{cross})}$ is the corresponding population-level probability of the opposite class.

Area under the perturbation curve. Each perturbation trajectory is summarised by its area under the curve (AUC), computed using the trapezoidal rule over the normalised perturbation fraction:

$$\text{AUC} = \int_0^1 \text{BA}(f)\,df \approx \sum_{k=0}^{N-1} \frac{\text{BA}(k) + \text{BA}(k+1)}{2} \cdot \frac{1}{N}, \tag{62}$$

where $N = 50$ for the segment-level benchmark and $N = 100$ for the timestep-level benchmark.

To quantify attribution faithfulness relative to the matched random-order baseline, we report

$$\Delta\text{AUC} = \text{AUC}_{\text{rand}} - \text{AUC}_{\text{attr}}. \tag{63}$$

Positive ΔAUC values indicate that attribution-guided perturbation degrades the model prediction more rapidly than the corresponding random-order baseline.

Bootstrap confidence intervals. Percentile bootstrap 95% confidence intervals for the mean ΔAUC are estimated from 2,000 bootstrap resamples of the per-sample values.

Wilcoxon signed-rank test. Statistical significance is assessed using a one-sided paired Wilcoxon signed-rank test with the alternative hypothesis

$$H_1 : \mathrm{AUC}_{\mathrm{attr}} < \mathrm{AUC}_{\mathrm{rand}}. \tag{64}$$

Bland–Altman analysis. Agreement between paired AUC measurements is assessed using Bland–Altman analysis. The mean difference (bias) and limits of agreement are computed as

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2}, \tag{65}$$

$$\mathrm{LoA}_{\mathrm{upper}} = \bar{d} + 1.96\, s_d, \qquad \mathrm{LoA}_{\mathrm{lower}} = \bar{d} - 1.96\, s_d. \tag{66}$$

Each Bland–Altman point represents one test sample, with the mean of the paired AUC values on the x-axis and their difference on the y-axis.

4

Results

In this section , we first evaluate the predictive performance of the two model architectures on the five-class PTB-XL classification task. Performance was assessed on the held-out test set over ten independent random initialisations using Accuracy, Macro F1, Macro AUC, and Macro Average Precision. Table 3 summarises the mean ± standard deviation performance across all random seeds, while Figure 25 shows the corresponding variability.

The results presented in Table 3 indicate that the two architectures achieved broadly comparable predictive performance. CNN achieved a higher mean Accuracy (0.785 vs. 0.766) and a marginally higher Macro Average Precision (0.743 vs. 0.742), while CNN–BiLSTM with Attention obtained a slightly higher Macro F1 score (0.672 vs. 0.656). Both models achieved nearly identical Macro AUC values (0.928 after rounding to three decimal places), although the unrounded means were 0.92825 for the CNN and 0.92787 for the CNN–BiLSTM with Attention.

Overall, the performance differences between the two architectures were small. This suggests that both models achieved comparable classification performance on the five-class task despite their different architectural designs and training strategies.

Model	Accuracy	Macro F1	Macro AUC	Macro Average Precision
CNN	0.785 ± 0.008	0.656 ± 0.022	0.928 ± 0.003	0.743 ± 0.010
CNN–BiLSTM	0.766 ± 0.011	0.672 ± 0.011	0.928 ± 0.003	0.742 ± 0.009

Table 3: Mean ± standard deviation classification performance on the five-class task across ten random seeds.

Figure 25 complements the summary statistics presented in Table 3 by illustrating the distribution of performance across the ten random seeds. The metric distributions overlap substantially for both architectures, further indicating that the observed performance differences are small. Macro AUC exhibits the least variability, suggesting stable ranking performance across different random initialisations. Slightly greater variability is observed for Macro F1 and Macro Average Precision, particularly for the CNN, although the distributions remain largely overlapping.

Overall, the box plots indicate that both architectures achieve comparable predictive

performance with consistent behaviour across different random seeds.

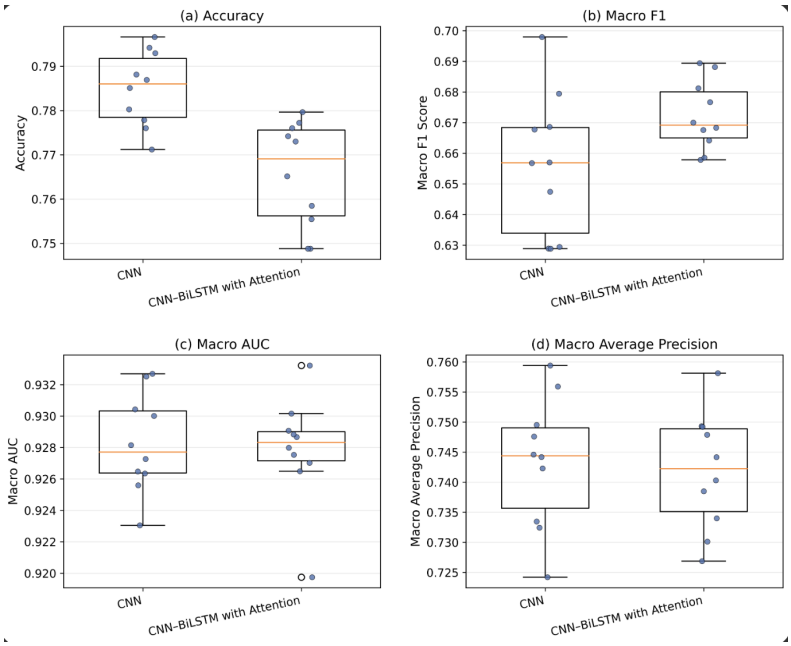
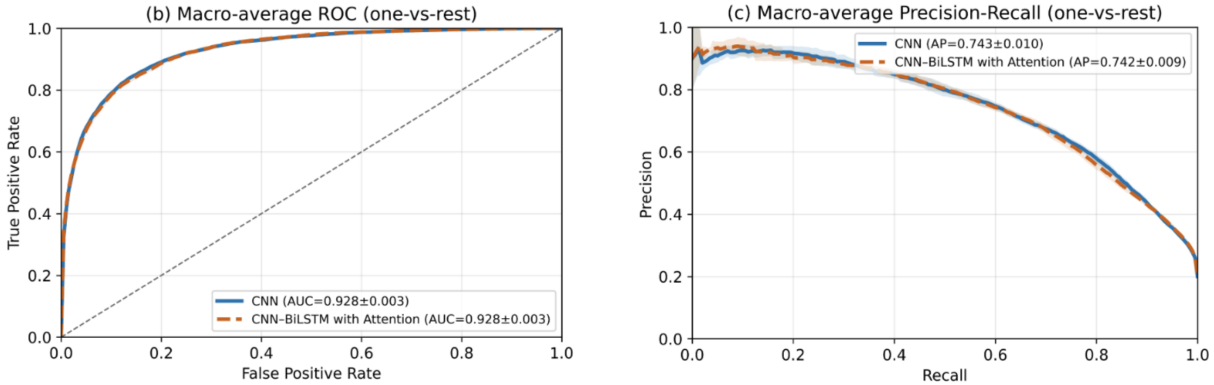


Figure 25: Distribution of Accuracy, Macro F1 score, Macro AUC, and Macro Average Precision for the CNN and CNN–BiLSTM with Attention models across ten random seeds. Each box shows the interquartile range and median; the ten individual seed values are overlaid as jittered points.

Figure 26(a,b) compares the macro-average ROC and Precision–Recall curves of the two architectures. Both models achieved nearly identical ROC-AUC (0.928 for both models) and macro Average Precision (0.743 for the CNN and 0.742 for the CNN–BiLSTM with Attention). The ROC curves almost completely overlap across the full operating range, while the Precision–Recall curves differ only slightly at some recall levels. Overall, both architectures exhibit very similar ranking performance on the five-class classification task.



(a) Macro-average ROC curves

(b) Macro-average Precision–Recall curves

Figure 26: Macro-average one-vs-rest ROC and Precision–Recall curves for the CNN and CNN–BiLSTM with Attention across ten random seeds.

4.1 Binary NORM-versus-MI Classification

The explainability experiments presented in the following sections are based on a binary NORM-versus-MI classification task. This simplified setting facilitates the interpretation of attribution maps and enables a more direct comparison of the different attribution methods.

The results presented in Table 4 show that the two architectures achieved highly comparable performance on the binary classification task. The CNN-BiLSTM with Attention achieved marginally higher mean Accuracy (0.912 vs. 0.907), Macro F1 (0.904 vs. 0.897), ROC-AUC (0.972 vs. 0.969), and Average Precision (0.960 vs. 0.956) than the CNN. However, the observed differences were small relative to the variability across random seeds, indicating that both models achieved highly comparable predictive performance on the held-out test set.

Model	Accuracy	Macro F1	ROC-AUC	Average Precision
CNN	0.907 ± 0.005	0.897 ± 0.007	0.969 ± 0.001	0.956 ± 0.002
CNN–BiLSTM	0.912 ± 0.005	0.904 ± 0.006	0.972 ± 0.002	0.960 ± 0.002

Table 4: Mean ± standard deviation classification performance on the binary NORM-versus-MI task across ten random seeds.

Figure 27 complements Table 4 by illustrating the distribution of performance across the ten random seeds. The CNN-BiLSTM with Attention consistently achieved slightly higher median values for all four metrics, while the distributions remained strongly overlapping. The narrow spread of the ROC-AUC and Average Precision values indicates stable performance across different random initialisations. Overall, both architectures exhibit highly comparable classification performance.

Figure 28 compares the ROC and Precision–Recall curves for the two architectures, with MI treated as the positive class. Both models achieved very similar ROC-AUC (0.969 for the CNN and 0.972 for the CNN-BiLSTM with Attention) and Average Precision (0.956 and 0.960, respectively). The ROC curves almost completely overlap across the full operating range, while only minor differences are visible in the Precision–Recall curves. These results further confirm that both architectures provide highly comparable discrimination performance on the binary NORM-versus-MI classification task.

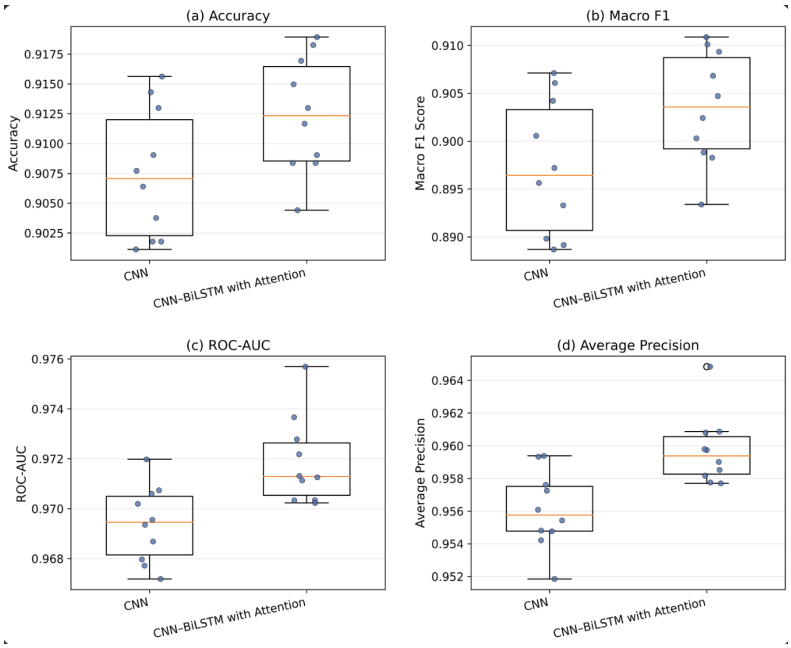


Figure 27: Distribution of Accuracy, Macro F1 score, ROC-AUC, and Average Precision for the CNN and CNN-BiLSTM with Attention models across ten random seeds. Each box shows the interquartile range and median; the ten individual seed values are overlaid as jittered points.

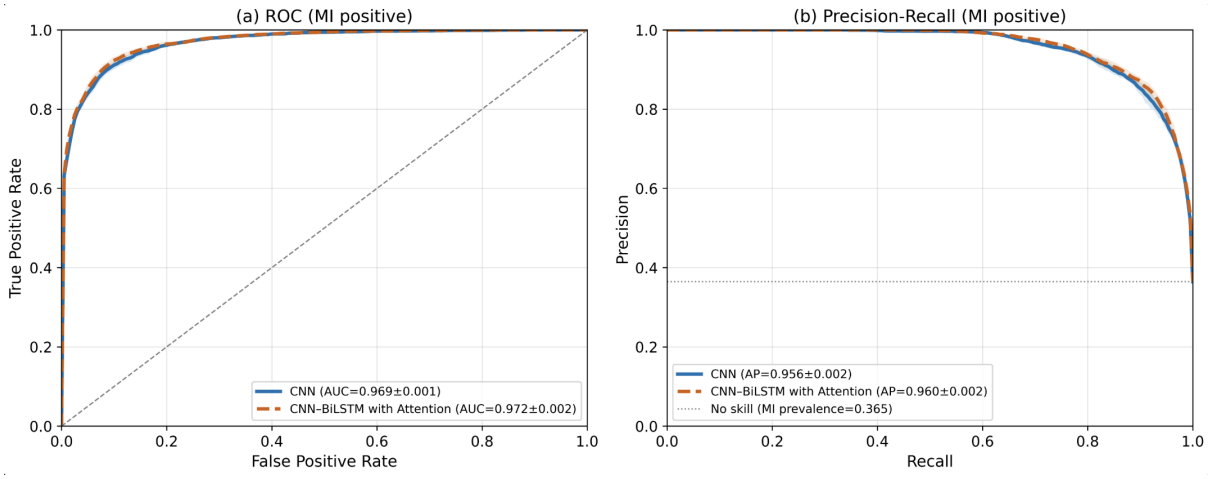


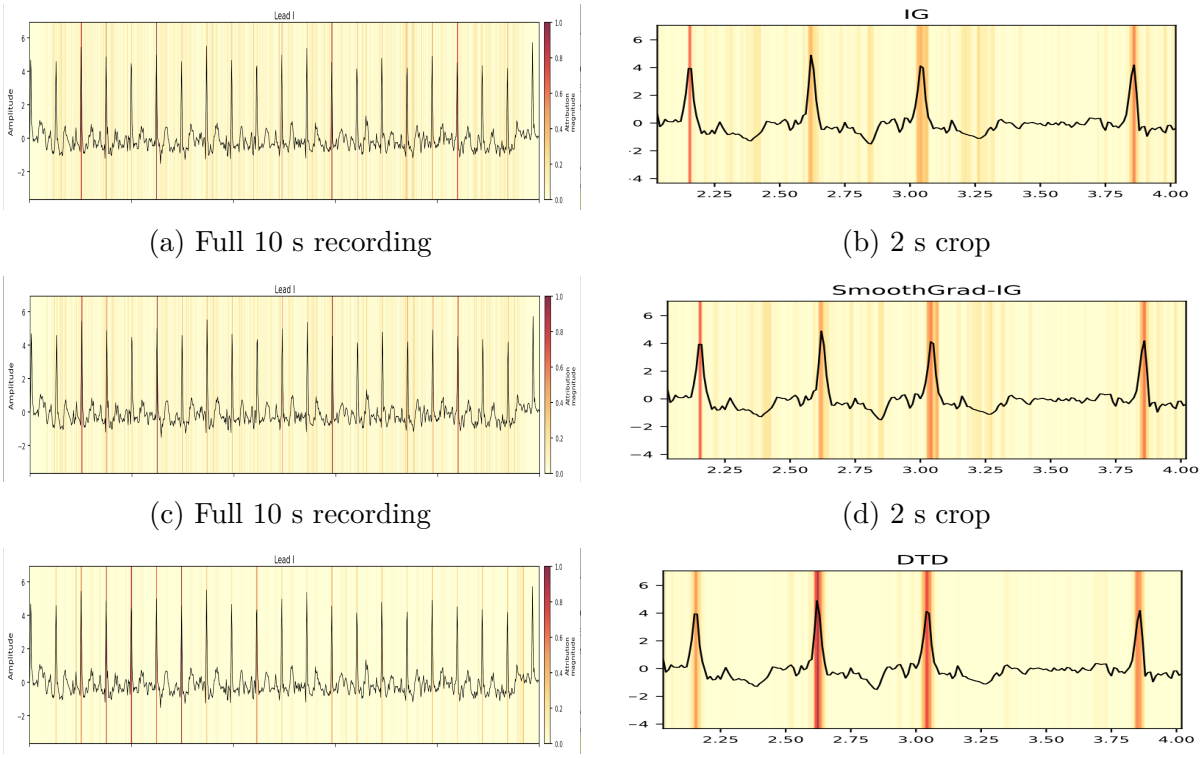
Figure 28: ROC (a) and Precision–Recall (b) curves for the CNN and CNN–BiLSTM models on the binary NORM-versus-MI classification task, aggregated across ten random seeds. MI is treated as the positive class. Shaded regions indicate ±1 standard deviation across seeds. The dotted line in the Precision–Recall curve denotes the no-skill baseline.

4.2 Qualitative Analysis of Attribution Heatmaps

Before performing the quantitative perturbation analysis, the attribution maps were examined qualitatively to understand where each attribution method assigned relevance on the ECG signal. A representative MI sample from the binary test set was selected and analysed using Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD). For consistency, the same ECG sample, lead, time axis, and colour scale were used for all visualisations, enabling a direct comparison across attribution methods and model architectures. The visualisations presented in this section and the subsequent perturbation analyses were generated using the same trained checkpoint for each architecture.

4.2.1 Comparison of Attribution Methods Within the CNN

Figure 29 compares the attribution patterns produced by the CNN for the same representative MI sample at two temporal scales. The left column shows the complete 10s Lead I recording, whereas the right column presents a fixed 2s interval from 2.03 to 4.03 s centred on a high-attribution QRS region. Across all three attribution methods, the highest relevance is consistently assigned to the QRS complexes, with the strongest attribution concentrated around the R-peak region, whereas the intervals between consecutive heartbeats generally receive substantially lower relevance. SmoothGrad-IG produces a broadly similar attribution pattern to IG while exhibiting visually smoother attribution maps. DTD also emphasizes the QRS complexes but produces a visually sparser and more localized relevance distribution, assigning comparatively less relevance to the surrounding waveform outside the QRS complexes. Compared with IG and SmoothGrad-IG, DTD appears to highlight the QRS complexes more selectively and to concentrate relevance more narrowly around the R-peak region. Overall, the three methods consistently identify the QRS complex as the dominant morphologic feature in this representative sample, while differing primarily in the spatial distribution of their attributed relevance. These qualitative observations describe the visual attribution patterns only and should not be interpreted as evidence of superior explanation faithfulness, which is evaluated separately using the perturbation benchmark.



(a) Full 10 s recording

(b) 2 s crop

(c) Full 10 s recording

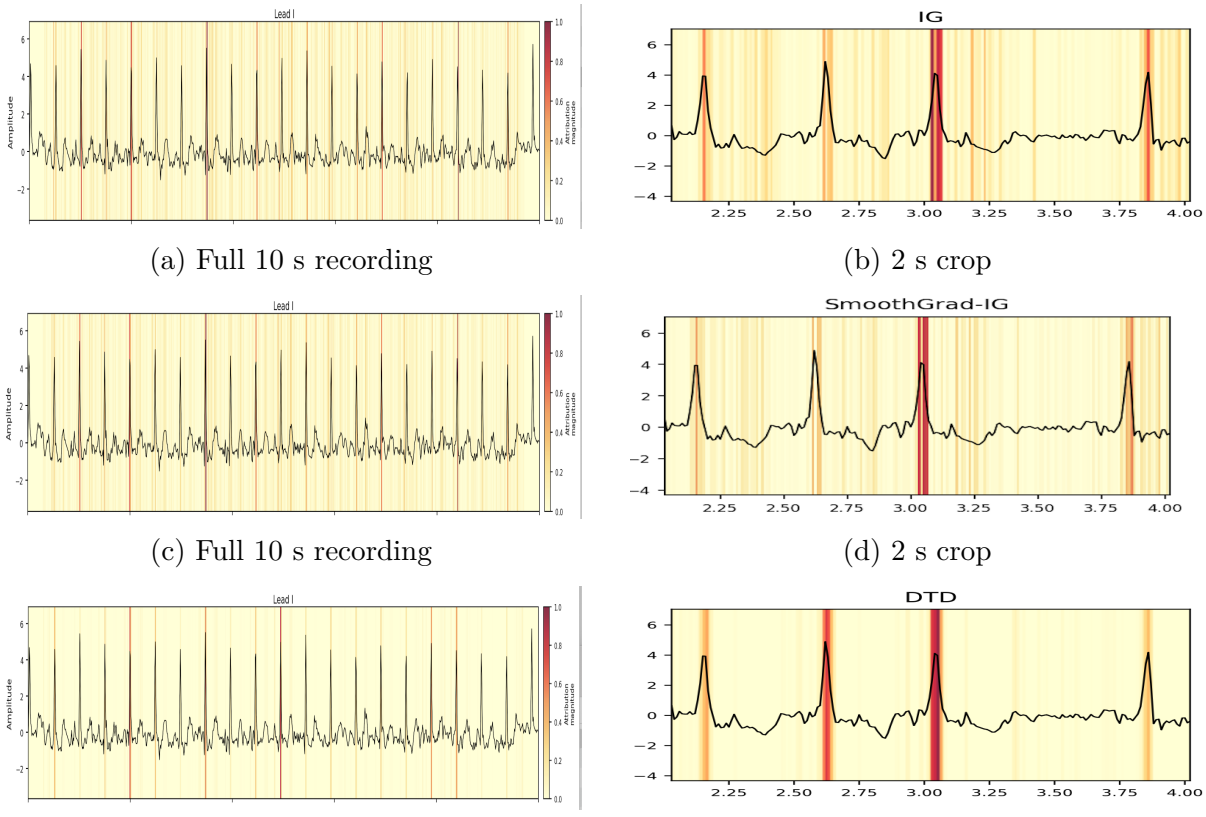
(d) 2 s crop

(e) Full 10 s recording

(f) 2 s crop

Figure 29: Comparison of full-recording and cropped attribution heatmaps for the CNN on the same representative MI sample

4.2.2 Comparison of Attribution Methods Within the CNN-BiLSTM



(a) Full 10 s recording

(b) 2 s crop

(c) Full 10 s recording

(d) 2 s crop

(e) Full 10 s recording

(f) 2 s crop

Figure 30: Comparison of full-recording and cropped attribution heatmaps for the CNN–BiLSTM on the same representative MI sample.

Figure 30 compares the attribution patterns produced by the CNN–BiLSTM with Attention for the same representative MI sample at two temporal scales. Across all three attribution methods, the highest relevance is consistently assigned to the QRS complexes, with the strongest attribution concentrated around the R-peak region, whereas the intervals between consecutive heartbeats generally receive substantially lower relevance. Integrated Gradients produces a relatively broader attribution pattern, with lower-intensity relevance extending into portions of the surrounding waveform between beats. SmoothGrad-IG identifies largely the same cardiac events while showing smoother transitions than IG. DTD also emphasizes the QRS complexes but produces a visually sparser and more localized relevance distribution, assigning comparatively less relevance to the surrounding waveform outside the QRS complexes. Compared with IG and SmoothGrad-IG, DTD appears to highlight the QRS complexes more selectively and to concentrate relevance more narrowly around the R-peak region. Overall, all three methods consistently identify the QRS complex as the dominant morphologic feature in this representative sample, while differing primarily in the spatial distribution of their attributed relevance. Similarly to the CNN, these qualitative observations describe only the visual attribution patterns

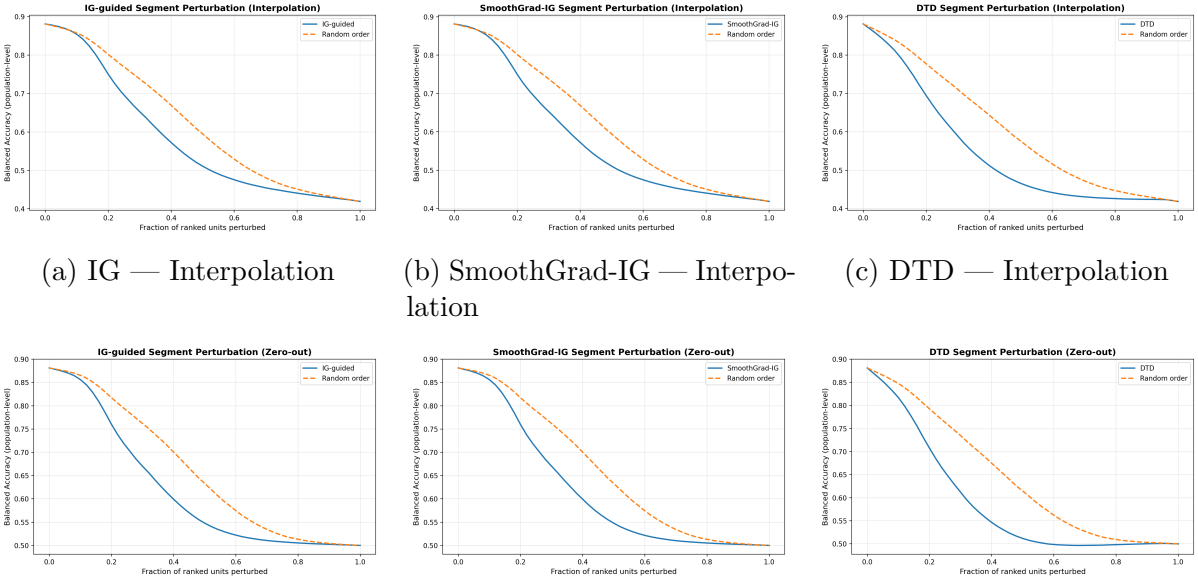
and should not be interpreted as evidence of superior explanation faithfulness.

Overall, both architectures exhibited highly similar qualitative attribution patterns. In both models, the highest relevance was consistently assigned to the QRS complexes, particularly around the R-peak region, and DTD produced visually sparser and more localized attribution maps than IG and SmoothGrad-IG. Nevertheless, subtle differences in the spatial distribution of relevance were observed. Compared with the CNN, the CNN–BiLSTM with Attention appeared to concentrate attribution more strongly on the central QRS complexes of the selected interval, whereas the CNN distributed relevance more evenly across the highlighted QRS complexes.

4.3 Quantitative Perturbation Analysis

The qualitative analysis presented in the previous section provides a visual comparison of the attribution maps. In this section, we evaluate the attribution methods quantitatively using perturbation analysis to assess whether the regions identified as important are truly relevant to the models' predictions. Population-level perturbation curves are presented for both architectures using interpolation- and zero-out-based perturbation strategies. Unless otherwise stated, all perturbation analyses in this chapter are performed on the entire binary test set.

4.4 CNN Perturbation Analysis



(a) IG — Interpolation (b) SmoothGrad-IG — Interpolation (c) DTD — Interpolation

(d) IG — Zero-out (e) SmoothGrad-IG — Zero-out (f) DTD — Zero-out

Figure 31: Population-level mean perturbation curves for the CNN model under interpolation and zero-out perturbation strategies. Each panel shows the population-level Balanced Accuracy as a function of the fraction of ranked segments perturbed (blue line), together with the mean curve obtained under random perturbation order (orange dashed line).

Figure 31 presents the population-level perturbation curves for the CNN model. Across all six panels, the attribution-guided perturbation curves remain below the corresponding random baseline for most of the perturbation range. This indicates that perturbing the regions identified by the attribution methods generally causes a larger degradation in model performance than perturbing randomly selected regions.

Comparing the two perturbation strategies, both interpolation- and zero-out-based perturbation exhibit a similar overall trend, with the largest reduction in performance occurring

during the early perturbation steps. However, interpolation generally reaches a lower final performance than zero-out, indicating a greater effect on the model predictions under the evaluated perturbation benchmark.

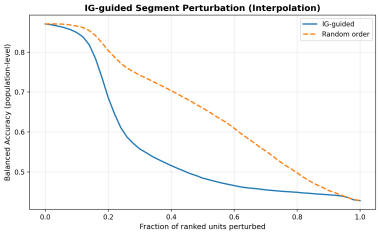
Across the three attribution methods, the perturbation curves follow highly similar trajectories. Integrated Gradients, SmoothGrad-IG, and Deep Taylor Decomposition (DTD) all remain clearly separated from the random baseline during the early and intermediate stages of perturbation. Although small differences in curve shape are visible, visual inspection alone is insufficient for drawing quantitative conclusions. Therefore, the following sections compare the methods using AUC and statistical analyses.

4.5 CNN-BiLSTM Perturbation Analysis

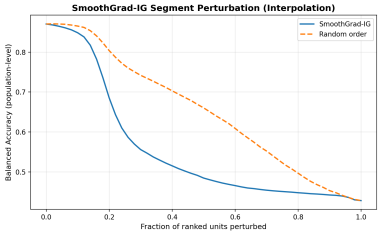
Figure 32 presents the population-level perturbation curves for the CNN–BiLSTM . Similar to the CNN, the attribution-guided perturbation curves remain below the corresponding random baseline for most of the perturbation range across all six panels. This indicates that perturbing the regions identified by the attribution methods generally causes a larger degradation in model performance than perturbing randomly selected regions.

Comparing the two perturbation strategies, both interpolation- and zero-out-based perturbation exhibit a similar overall trend, with the largest reduction in performance occurring during the early perturbation steps. As observed for the CNN, interpolation generally reaches a lower final performance than zero-out, indicating a greater effect on the model predictions under the evaluated perturbation benchmark.

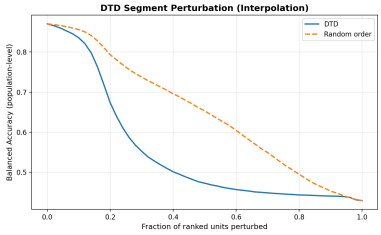
Across the three attribution methods, the perturbation curves follow highly similar trajectories. Integrated Gradients, SmoothGrad-IG, and Deep Taylor Decomposition (DTD) all remain clearly separated from the random baseline during the early and intermediate stages of perturbation. Although small differences in curve shape are visible, no consistent qualitative differences can be established from visual inspection alone. A more detailed quantitative comparison based on AUC and statistical analysis is presented in the following sections.



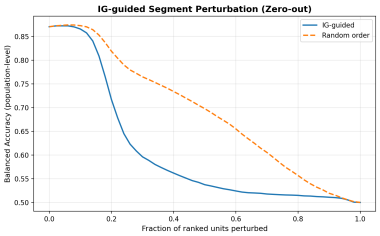
(a) IG — Interpolation



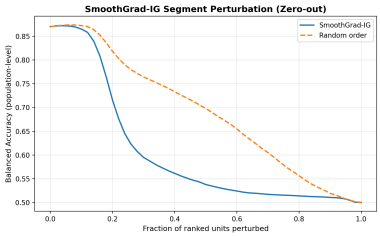
(b) SmoothGrad-IG — Interpolation



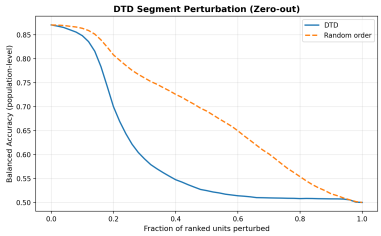
(c) DTD — Interpolation



(d) IG — Zero-out



(e) SmoothGrad-IG — Zero-out



(f) DTD — Zero-out

Figure 32: Population-level mean perturbation curves for the **CNN-BiLSTM** model under interpolation and zero-out perturbation strategies. Each panel shows the population-level Balanced Accuracy as a function of the fraction of ranked segments perturbed (blue line), together with the mean curve obtained under random perturbation order (orange dashed line)

4.5.1 Cross-Architecture and Cross-Method Comparison

This figure brings together the mean segment-level perturbation curves for both model architectures, all three attribution methods, and both perturbation strategies in a single four-panel layout, enabling direct visual comparison across models and experimental settings. Across all four panels, the attribution-guided curves remain below the corresponding random baseline for most of the perturbation range, indicating that perturbing the regions identified by the attribution methods generally causes a larger degradation in model performance than perturbing randomly selected regions. DTD tends to exhibit the greatest separation from the random baseline, whereas Integrated Gradients (IG) and SmoothGrad-IG follow highly similar trajectories.

More precisely (see Figure 33), within the CNN, DTD produces a steeper reduction in performance than IG and SmoothGrad-IG under both perturbation strategies, whereas the latter two methods remain nearly indistinguishable.

In the CNN-BiLSTM with Attention, the three attribution methods follow even more similar trajectories.

Compared with the CNN, however, the CNN-BiLSTM with Attention exhibits a steeper early decline in performance, suggesting greater sensitivity to perturbing the highest-ranked

ECG segments under the evaluated perturbation benchmark.

Whether these visual differences are statistically meaningful is examined in the following section.

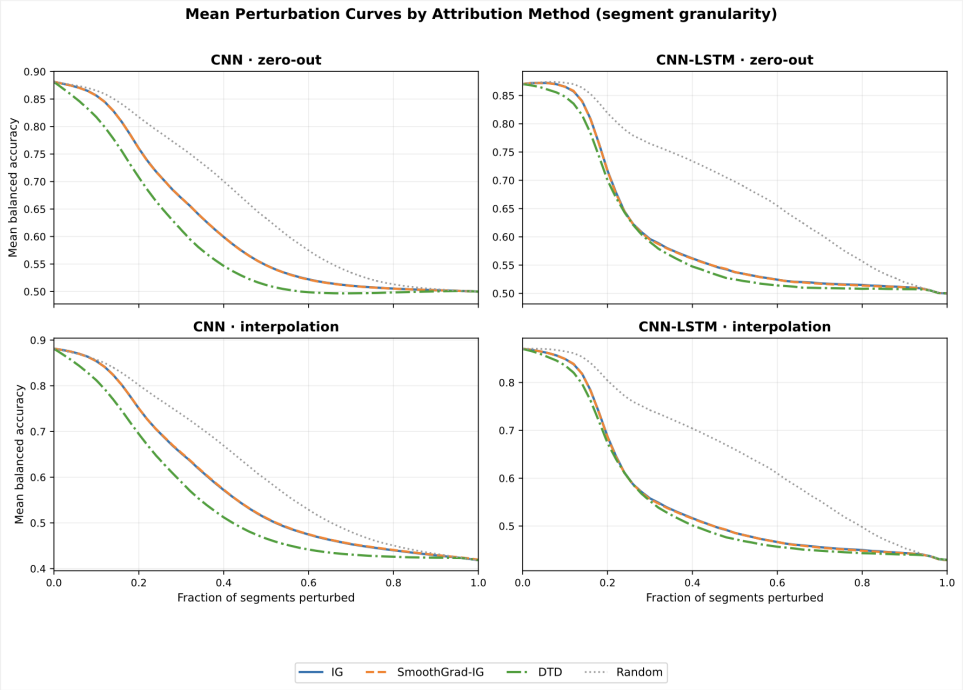


Figure 33: Summary of the population-level segment perturbation curves for both architectures, attribution methods, and perturbation strategies. Attribution-guided curves are compared with the random baseline.

4.5.2 Population-level AUC and Statistical Comparison

Based on the qualitative comparison of the perturbation curves, this section quantitatively evaluates attribution faithfulness using population-level AUC and statistical analysis. The objective is to determine whether the observed differences between attribution methods, model architectures, and perturbation strategies are supported by statistical evidence.

Table 5 summarises the population-level perturbation results for all model–method–strategy combinations. Across both architectures and perturbation strategies, all attribution methods outperform the corresponding random baseline, yielding positive ΔAUC values together with highly significant paired Wilcoxon signed-rank test results (all $p < 10^{-15}$). In addition, the 95% bootstrap confidence intervals for ΔAUC remain entirely above zero, providing further support that attribution-guided perturbation results in larger performance degradation than random perturbation under the evaluated benchmark.

Within the CNN, Integrated Gradients (IG) and SmoothGrad-IG achieve nearly identical results under both perturbation strategies, whereas Deep Taylor Decomposition (DTD)

consistently produces the largest ΔAUC values. The same overall pattern is observed for the CNN–BiLSTM with Attention, where DTD again achieves the largest ΔAUC, followed closely by IG and SmoothGrad-IG. Across all attribution methods, the CNN–BiLSTM with Attention generally achieves larger ΔAUC values than the CNN, suggesting higher attribution faithfulness under the evaluated perturbation benchmark. Finally, interpolation generally yields slightly larger ΔAUC values than zero-out, consistent with the larger performance degradation observed in the perturbation curves. An exception is observed for the CNN under Integrated Gradients and SmoothGrad-IG, where zero-out yields marginally larger ΔAUC values.

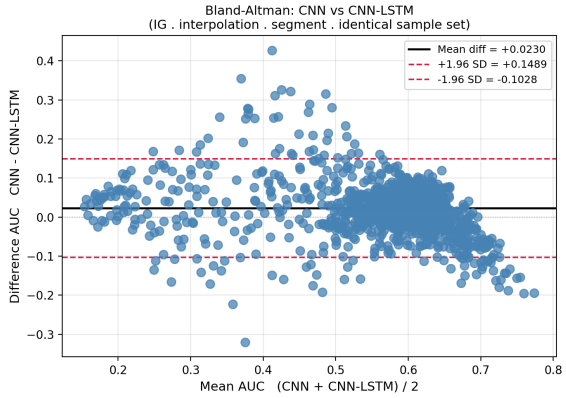
Model	**Method**	**Strategy**	**Method AUC**	**Random AUC**	**ΔAUC**	**95% CI (Δ)**	**Win rate**	***p*-value**
CNN	IG	Zero-out	0.6165	0.6594	0.0429	[0.0405, 0.0454]	86.8%	2.29×10^{-163}
CNN	IG	Interpolation	0.5795	0.6207	0.0413	[0.0385, 0.0439]	77.1%	2.29×10^{-123}
CNN	SmoothGrad-IG	Zero-out	0.6163	0.6594	0.0430	[0.0406, 0.0455]	86.7%	4.32×10^{-163}
CNN	SmoothGrad-IG	Interpolation	0.5790	0.6202	0.0412	[0.0384, 0.0440]	76.8%	3.19×10^{-123}
CNN	DTD	Zero-out	0.5875	0.6454	0.0579	[0.0555, 0.0601]	97.0%	1.77×10^{-238}
CNN	DTD	Interpolation	0.5442	0.6061	0.0620	[0.0595, 0.0644]	93.1%	1.29×10^{-225}
CNN-BiLSTM	IG	Zero-out	0.6039	0.6911	0.0873	[0.0823, 0.0926]	80.7%	4.07×10^{-148}
CNN-BiLSTM	IG	Interpolation	0.5564	0.6542	0.0977	[0.0932, 0.1025]	87.2%	1.62×10^{-194}
CNN-BiLSTM	SmoothGrad-IG	Zero-out	0.6034	0.6910	0.0876	[0.0826, 0.0928]	80.6%	2.19×10^{-148}
CNN-BiLSTM	SmoothGrad-IG	Interpolation	0.5557	0.6538	0.0981	[0.0936, 0.1028]	86.5%	6.63×10^{-193}
CNN-BiLSTM	DTD	Zero-out	0.5942	0.6857	0.0915	[0.0866, 0.0964]	85.8%	7.91×10^{-174}
CNN-BiLSTM	DTD	Interpolation	0.5480	0.6489	0.1009	[0.0964, 0.1054]	90.1%	1.46×10^{-208}

Table 5: Population-level AUC summary for segment-level perturbation across all experimental settings

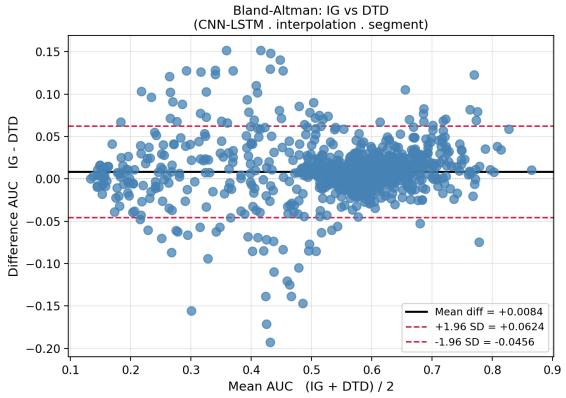
4.6 Bland–Altman Agreement Analysis

To complement the population-level perturbation analysis, Bland–Altman analysis was performed to assess sample-level agreement between model architectures, attribution methods, and perturbation strategies. Specifically, the analysis evaluated the agreement between the perturbation AUC measurements obtained under the different experimental conditions. The comparison between interpolation- and zero-out-based perturbation was restricted to MI samples, as the interpolation perturbation was not applied to NORM recordings.

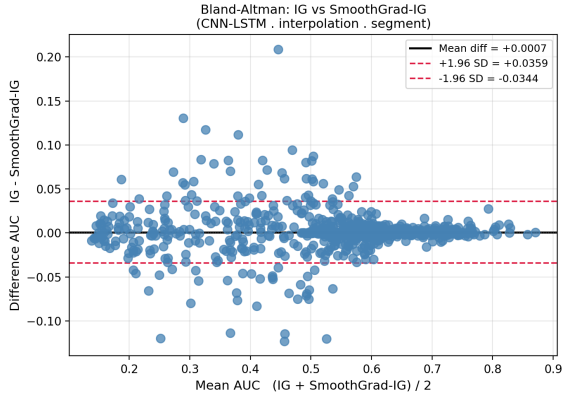
Figure 34 summarises the agreement between the different experimental conditions. The closest agreement is observed between Integrated Gradients (IG) and SmoothGrad-IG, whereas comparisons involving Deep Taylor Decomposition (DTD) show somewhat larger differences. A moderate level of disagreement is also observed between the CNN and the CNN–BiLSTM with Attention. The largest disagreement is observed between the interpolation- and zero-out-based perturbation strategies, suggesting that the perturbation strategy contributes more to variation in sample-level perturbation AUC than the attribution method under the evaluated benchmark.



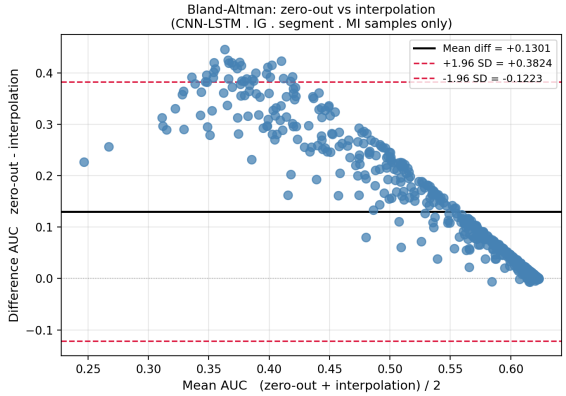
(a) CNN vs. CNN-BiLSTM (IG, interpolation, segment).



(b) IG vs. DTD (CNN-BiLSTM, interpolation, segment).



(c) IG vs. SmoothGrad-IG (CNN-BiLSTM, interpolation, segment).



(d) Zero-out vs. interpolation (CNN-BiLSTM, IG, segment; MI samples only).

Figure 34: Bland Altman agreement plots for four representative pairwise comparisons of perturbation AUC values. IG and SmoothGrad-IG show the closest agreement, whereas the largest disagreement is observed between the zero-out and interpolation perturbation strategies

5

Discussion

This thesis examined the explainability of deep learning models for ECG-based myocardial infarction detection by combining qualitative attribution analysis with a quantitative perturbation benchmark. Two neural network architectures, namely a CNN and a CNN–BiLSTM with Attention, were evaluated using Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD). In contrast to studies that rely primarily on visual inspection of attribution maps, this work complemented qualitative observations with perturbation-based faithfulness evaluation, statistical hypothesis testing, and Bland–Altman agreement analysis.

The classification experiments demonstrated that both architectures achieved strong predictive performance on both the five-class PTB-XL classification task and the binary NORM-versus-MI classification task. Across both tasks, the two architectures achieved highly comparable predictive performance. While the relative advantage differed slightly between the five-class and binary settings, the observed performance differences remained small and were unlikely to influence the subsequent explainability analyses.

The qualitative attribution analysis revealed several consistent patterns across all three attribution methods. Regardless of the underlying architecture, the attribution maps concentrated most of their relevance around the QRS complexes, particularly near the R-peaks. From a clinical perspective, this observation is consistent with the known importance of ventricular depolarisation abnormalities in myocardial infarction. Also Integrated Gradients and SmoothGrad-IG produced broadly similar attribution patterns, with SmoothGrad-IG exhibiting visually smoother attribution maps. DTD produced visually sparser and more localised relevance distributions while remaining concentrated around the QRS complexes. Although minor qualitative differences in the spatial distribution of relevance were observed, these differences should be interpreted with caution. Visual inspection alone does not provide evidence of explanation quality; therefore, explanation faithfulness should be assessed using quantitative evaluation, such as the perturbation benchmark employed in this study.

One of the main contributions of this thesis is the use of perturbation analysis to evaluate attribution faithfulness beyond visual inspection. The perturbation experiments progressively replaced the highest-ranked ECG segments and measured the resulting degradation in predictive performance. Across both architectures, attribution-guided

perturbation generally produced a larger degradation than the corresponding random baseline, indicating that the attribution methods identified ECG regions that had a greater influence on model predictions than randomly selected regions.

Comparisons across the three attribution methods revealed relatively modest but consistent differences. Integrated Gradients and SmoothGrad-IG produced highly similar perturbation curves and AUC values, consistent with the design of SmoothGrad-IG, which averages multiple noisy attribution maps while largely preserving the underlying attribution pattern. Deep Taylor Decomposition generally achieved the lowest perturbation AUC values and the largest ΔAUC values, observed across both perturbation strategies and both model architectures. Overall, these results suggest that DTD achieved the highest attribution faithfulness among the evaluated methods. It should be noted that DTD could be applied directly to the CNN, whereas the CNN–BiLSTM with Attention required a hybrid implementation because exact relevance propagation is not fully supported through recurrent layers; the resulting DTD maps were nonetheless broadly consistent with the other methods and achieved competitive performance, suggesting a reasonable approximation.

The comparison between the two architectures provides additional insight into how model design influences explainability. Although both models achieved similar classification performance, the CNN–BiLSTM with Attention generally produced larger perturbation effects than the CNN. Interestingly, this difference is not immediately apparent from the attribution maps alone, as the CNN–BiLSTM with Attention often produced visually broader relevance distributions; the perturbation analysis showed that perturbing its highest-ranked regions resulted in larger changes in predictive performance than for the CNN, consistent with greater attribution faithfulness. These findings highlight the importance of complementing qualitative attribution visualisations with quantitative perturbation-based evaluation when comparing explainability across model architectures.

Across most experimental settings, interpolation generally produced larger degradation than zero-out masking. One possible explanation is that interpolation replaces important ECG segments with physiologically plausible healthy waveforms, removing pathological information while preserving signal continuity, whereas zero-out masking introduces an artificial flat baseline that may leave information in neighbouring regions relatively unaffected. The results are therefore consistent with interpolation providing a more realistic perturbation strategy for ECG signals. These observations were further supported statistically: bootstrap confidence intervals remained above zero for all configurations, and paired Wilcoxon signed-rank tests indicated statistically significant differences between attribution-guided and random perturbation, suggesting that the observed differences are unlikely to be explained by sampling variability alone. The Bland–Altman analysis showed close agreement between IG and SmoothGrad-IG, and somewhat larger differences involving DTD; the largest disagreement was observed between interpolation- and zero-out-based perturbation, indicating that the choice of perturbation strategy contributed

more to variation in the measured faithfulness than the choice of attribution method.

Despite these encouraging findings, this thesis has several limitations that should be acknowledged. First, all experiments were conducted using the PTB-XL dataset, so the extent to which the findings generalise to other ECG datasets or clinical populations remains uncertain. Second, the explainability analysis focused on the binary NORM-versus-MI classification task, whereas more complex multi-class settings may present additional challenges. Third, the study evaluated three attribution methods : two gradient-based (IG and SmoothGrad-IG) and one propagation-based (DTD); other techniques, including perturbation-based and concept-based approaches, could provide complementary insights. Finally, the interpolation strategy relied on synthetically generated healthy ECG signals; patient-specific healthy recordings may offer a more realistic basis for perturbation in future studies.

6

Conclusion

This thesis investigated the explainability of deep learning models for electrocardiogram (ECG) classification, with a particular focus on myocardial infarction (MI) detection. Two neural network architectures, namely a CNN and a CNN–BiLSTM with Attention, were evaluated using three attribution methods: Integrated Gradients (IG), SmoothGrad-IG, and Deep Taylor Decomposition (DTD). Rather than relying solely on qualitative inspection of attribution maps, this work introduced an attribution-guided perturbation benchmark combined with statistical analyses to evaluate attribution faithfulness quantitatively.

Based on our results, there are several consistent findings. First, attribution-guided perturbation produced larger degradation in predictive performance than random perturbation across the evaluated settings, suggesting that the highlighted ECG regions were informative for the models' predictions. Second, Deep Taylor Decomposition generally achieved the highest attribution faithfulness under the evaluated perturbation benchmark, whereas Integrated Gradients and SmoothGrad-IG produced highly consistent results across both architectures. Third, although the CNN and CNN–BiLSTM with Attention achieved comparable classification performance, the CNN–BiLSTM with Attention generally exhibited larger perturbation effects, indicating higher attribution faithfulness despite producing appeared visually broader relevance maps. Finally, interpolation-based perturbation generally resulted in greater performance degradation than zero-out masking, while Bland–Altman analysis suggested that the choice of perturbation strategy contributed more to variation in measured attribution faithfulness than the choice of attribution method.

Taken together, these findings suggest that explainability assessment should not rely solely on visual inspection of attribution maps. Combining qualitative analysis with perturbation-based evaluation and statistical analyses provides a more comprehensive assessment of attribution faithfulness. The proposed benchmark offers a practical and reproducible framework for evaluating attribution methods in ECG analysis and may support the development of more transparent and trustworthy deep learning systems for clinical decision support.

Several directions remain for future work. The proposed framework could be extended by evaluating additional attribution methods, investigating other cardiovascular diseases and

multi-class diagnostic settings, and validating the benchmark on external clinical datasets. Furthermore, incorporating patient-specific reference signals or clinically annotated ECG landmarks may further improve the physiological realism of perturbation-based explanation evaluation.

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A

Additional Classification Results

Model	True class	Pred. NORM	Pred. MI
CNN	NORM	911	53
	MI	85	468
CNN–BiLSTM–Attn	NORM	928	36
	MI	103	450

Table 6: Confusion matrices for the binary NORM-versus-MI classification task.

Table 6 shows Confusion matrices for the CNN and CNN–BiLSTM with Attention models on the held-out binary NORM-versus-MI test set (1517 samples: 964 NORM, 553 MI), for the seed-42 models. Rows are the true class and columns the predicted class. The matrices complement the aggregate metrics in Table 4 by showing the error breakdown; in particular, the CNN misses 85 of 553 MI cases (sensitivity ≈ 0.85) and the CNN–BiLSTM 103 (sensitivity ≈ 0.81).

B

Completeness Check Results

Table 7 reports the Integrated Gradients completeness relative error ϵ_{rel} for the two evaluated architectures using a zero baseline. The median relative errors of 0.52 % for the CNN model and 0.80 % for the CNN-BiLSTM-Attention model indicate good numerical agreement with the completeness property. Using m = 50 integration steps, all CNN samples and 95 % of CNN-BiLSTM-Attention samples remained below the 5 % threshold, confirming that the numerical approximation was sufficiently accurate for the experiments. As expected, the completeness check is only applicable to Integrated Gradients and was therefore not performed for SmoothGrad-IG.

Model	Median ϵ_{rel}	Max ϵ_{rel}	% < 5%
CNN	0.52%	3.98%	100%
CNN-BiLSTM-Attn	0.80%	14.36%	95%

Table 7: Integrated Gradients completeness check using m = 50 integration steps and a zero baseline on 20 MI test samples. Median and maximum relative completeness errors are reported together with the percentage of samples whose relative error is below 5 %

C

Additional Bland–Altman Agreement Results

Bland–Altman analysis was additionally performed to quantify the agreement between representative pairs of attribution methods, perturbation strategies, and model architectures. The corresponding summary statistics are provided in Table 8. The main Bland–Altman results and representative plots are presented in Chapter 4.

Comparison (A vs. B)	Mean diff	LoA lower	LoA upper	LoA width
IG vs. SmoothGrad-IG	+0.0007	-0.0344	+0.0359	0.070
IG vs. DTD	+0.0084	-0.0456	+0.0624	0.108
CNN vs. CNN–LSTM with Attention	+0.0230	-0.1028	+0.1489	0.252
Zero-out vs. Interpolation (MI only)	+0.1301	-0.1223	+0.3824	0.505

Table 8: Table 8 Bland–Altman agreement statistics for the four representative comparisons shown in Figure 34 (n = 1517). Mean diff denotes the mean difference $\bar{d}$, and LoA denotes the 95% limits of agreement.

D

Additional Bland-Altman

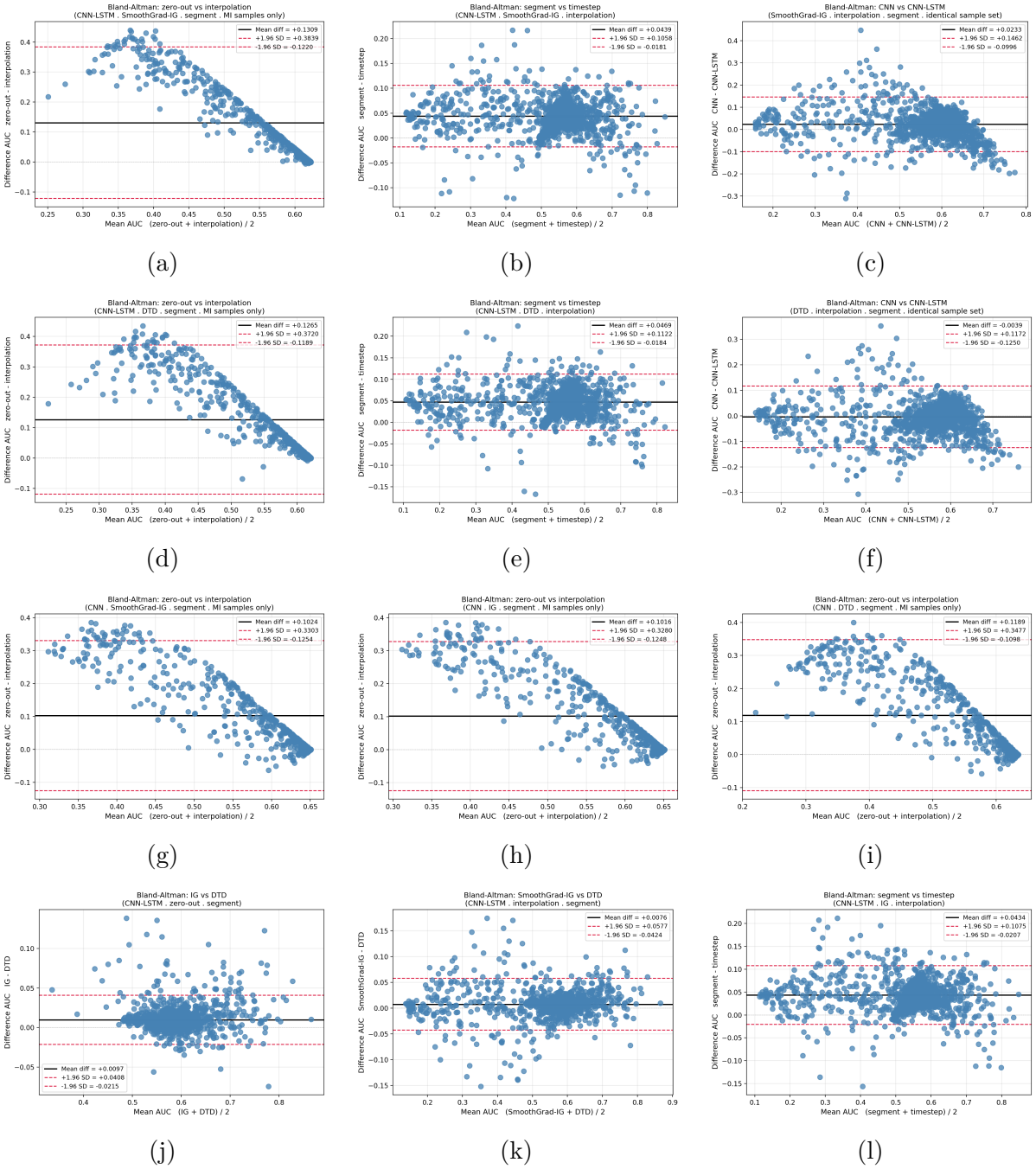


Figure 35: Bland–Altman analyses of sample-level perturbation AUC values for the pairwise comparisons across attribution methods, model architectures, and perturbation strategies. Each panel shows the mean difference and the corresponding 95% limits of agreement. The analyses include MI samples only.

E

Mathematical Derivation of IG Completeness

We provide a brief derivation of the completeness axiom for Integrated Gradients. Given a model F, input $\mathbf{x}$, and baseline $\tilde{\mathbf{x}}$, the IG attribution for feature i is:

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - \tilde{x}_i) \int_0^1 \frac{\partial F(\tilde{\mathbf{x}} + \alpha(\mathbf{x} - \tilde{\mathbf{x}}))}{\partial x_i} \, d\alpha \tag{67}$$

Summing over all features:

$$\begin{aligned}
\sum_i \mathrm{IG}_i(\mathbf{x}) &= \sum_i (x_i - \tilde{x}_i) \int_0^1 \frac{\partial F(\tilde{\mathbf{x}} + \alpha(\mathbf{x} - \tilde{\mathbf{x}}))}{\partial x_i} \, d\alpha && (68)\\
&= \int_0^1 \sum_i (x_i - \tilde{x}_i) \frac{\partial F(\tilde{\mathbf{x}} + \alpha(\mathbf{x} - \tilde{\mathbf{x}}))}{\partial x_i} \, d\alpha && (69)\\
&= \int_0^1 \nabla F(\tilde{\mathbf{x}} + \alpha(\mathbf{x} - \tilde{\mathbf{x}}))^\top (\mathbf{x} - \tilde{\mathbf{x}}) \, d\alpha && (70)\\
&= \int_0^1 \frac{d}{d\alpha} F(\tilde{\mathbf{x}} + \alpha(\mathbf{x} - \tilde{\mathbf{x}})) \, d\alpha && (71)\\
&= F(\mathbf{x}) - F(\tilde{\mathbf{x}}) && (72)
\end{aligned}$$

F

Additional Timestep-Level perturbation Results

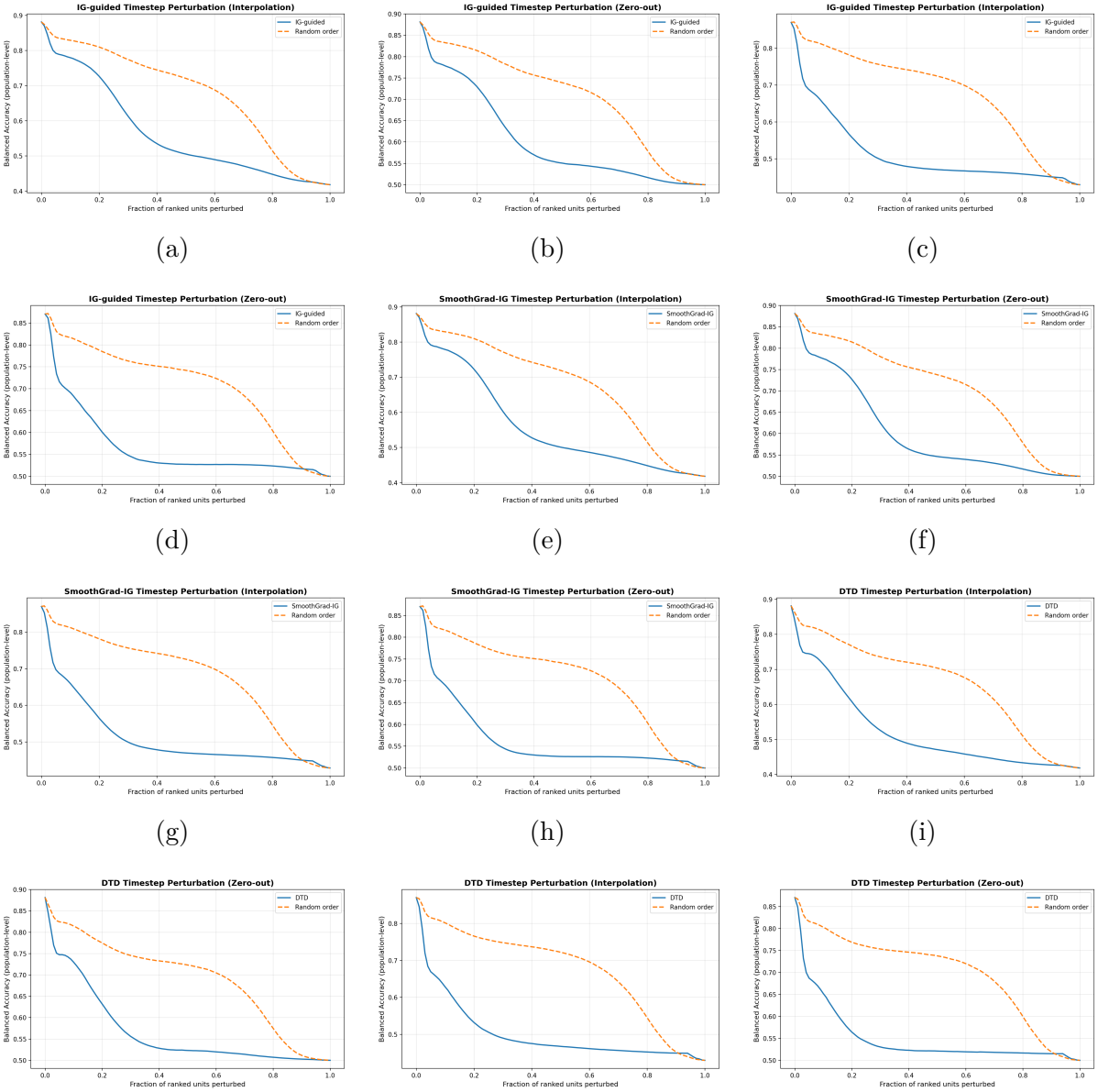


Figure 36: Timestep-level perturbation curves for the CNN and CNN–BiLSTM with Attention. The panels compare Integrated Gradients, SmoothGrad-IG, and Deep Taylor Decomposition under interpolation and zero-out perturbation. For each configuration, population-level balanced accuracy is shown as progressively larger fractions of the highest-ranked timesteps are perturbed, alongside the matched random-order baseline

G

Relevance Conservation in Deep Taylor Decomposition

This appendix provides the mathematical proof underlying the relevance-conservation property summarized in section 2.9 for the specific Deep Taylor Decomposition (DTD) variant used in this thesis, namely the z^+ rule applied layer-by-layer through the convolutional network. We first recall the propagation rule as implemented, prove local conservation for each layer type, and then chain these results into global conservation. We close with the two practical caveats (the numerical stabiliser and the CNN–BiLSTM hybrid) under which conservation holds only approximately.

G.1 Setup and the z^+ propagation rule

Consider a layer that maps a non-negative input vector $\mathbf{x} = (x_i)_i$ to pre-activations $\left(z_j\right)_j$ through weights w_{ij},

$$z_j \;=\; \sum_i x_i\, w_{ij} + b_j,$$

followed by a ReLU nonlinearity. Writing $x_i^+ = \max(x_i, 0)$ and $w_{ij}^+ = \max(w_{ij}, 0)$, the z^+ rule redistributes the relevance R_j of an upper-layer neuron j to a lower-layer neuron i as

$$R_{i\leftarrow j} \;=\; \frac{x_i^+\, w_{ij}^+}{\sum_{i'} x_{i'}^+\, w_{i'j}^+}\, R_j, \qquad R_i \;=\; \sum_j R_{i\leftarrow j}. \tag{73}$$

This is the closed form of the first-order Taylor expansion in Equation (23) evaluated at the root point $\tilde{\mathbf{x}}$ lying in the positive orthant, where $\partial h_j/\partial x_i$ selects the positive weights w_{ij}^+ and the displacement $(x_i - \tilde{x}_i)$ is proportional to x_i^+. Equation (73) is exactly the operation implemented for the linear and Conv1d layers, with $z_j^+ = \sum_i x_i^+ w_{ij}^+$ the positive pre-activation.

G.2 Local conservation for linear and convolutional layers

Lemma 1 (Layer-wise conservation)**.** *For the z^+ rule* (73)*, the total relevance is preserved across the layer:*

$$\sum_i R_i \;=\; \sum_j R_j.$$

Proof. Summing (73) over the lower-layer index i and exchanging the finite sums,

$$\sum_i R_i = \sum_i \sum_j \frac{x_i^+ w_{ij}^+}{z_j^+} R_j = \sum_j \frac{R_j}{z_j^+} \underbrace{\sum_i x_i^+ w_{ij}^+}_{=z_j^+} = \sum_j R_j.$$

The interchange is valid because both sums are finite, and $z_j^+ > 0$ whenever neuron j receives relevance. □

Because a convolution is a linear map with shared weights, the same argument applies verbatim to the Conv1d layers: the transposed convolution in the implementation is precisely the adjoint operator $\sum_j w_{ij}^+(\cdot)$, so the messages $R_{i\leftarrow j}$ again sum to R_j.

G.3 Conservation for pooling and parameter-free layers

Max-pooling. Each output position copies the value of a single winning input position $i^\star(j) = \arg\max_{i\in\mathcal{P}_j} x_i$, where $\mathcal{P}_j$ is the pooling window. Relevance is routed entirely to that position, $R_i = \sum_{j:\, i^\star(j)=i} R_j$ and 0 otherwise (the `max_unpool` operation). Hence $\sum_i R_i = \sum_j R_j$ exactly.

Adaptive average pooling. The global pooling layer is inverted with the z^+ redistribution $R_i = \dfrac{x_i^+}{\sum_{i'} x_{i'}^+} R$, whose weights sum to one, so $\sum_i R_i = R$.

ReLU and BatchNorm. ReLU carries no parameters; relevance is assigned to the pre-activation at the same index, which is conservative. In evaluation mode BatchNorm is a fixed affine pass-through and is treated as transparent, so it does not alter the relevance total.

G.4 Global conservation

Let L index the output layer. DTD is initialised by placing the network output for the target class c on the corresponding output neuron,

$$R_k^{(L)} = f_c(\mathbf{x})\,\delta_{kc}, \qquad \text{so} \qquad \sum_k R_k^{(L)} = f_c(\mathbf{x}).$$

By the Lemma of G.2 and the results of G.3, every layer l satisfies $\sum_i R_i^{(l)} = \sum_j R_j^{(l+1)}$ (Equation (20)). Applying this identity inductively from layer L down to the input layer $l = 0$ gives

$$\sum_i R_i^{(0)} \;=\; \sum_k R_k^{(L)} \;=\; f_c(\mathbf{x}), \tag{74}$$

which is the global relevance-conservation property of Equation (19): the input relevances sum to the model output for the explained class.

G.5 Practical caveats

Numerical stabiliser. The implementation replaces z_j^+ by $z_j^+ + \varepsilon$ with $\varepsilon = 10^{-9}$ to avoid division by zero. This makes conservation hold up to a bounded leakage,

$$\sum_i R_i = \sum_j \frac{z_j^+}{z_j^+ + \varepsilon}\, R_j \;\le\; \sum_j R_j,$$

with equality in the limit $\varepsilon \to 0$. For the activation magnitudes encountered in practice the relative leakage is negligible.

CNN–BiLSTM hybrid. For the CNN–BiLSTM with Attention model, the recurrent gates admit no conservation-preserving root point, so relevance is propagated through the recurrent and attention block by the gradient $\times$ input approximation and the z^+ rule is applied only within the convolutional sub-network. Consequently, Equation (74) holds exactly (up to ε) only for the purely convolutional CNN model; for the CNN–BiLSTM it holds within the convolutional part, while the recurrent/attention stage is treated as an attribution-carrying but non-conservative interface.